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Gaze Mapping

This Help Center article explains how to use 'gaze mapping' for mapping dynamic eye tracking data onto a static image.

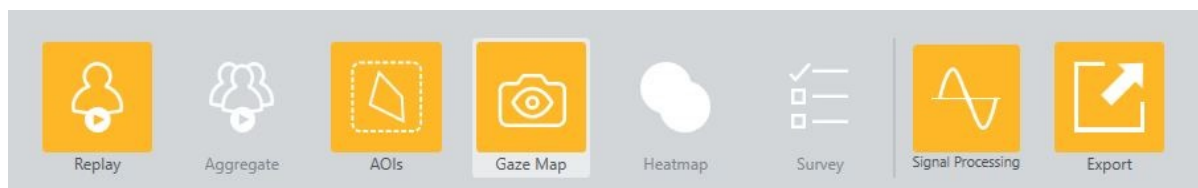
Gaze mapping is a tool for mapping *dynamic* eye tracking data, that is, eye tracking data recorded in a *dynamic* environment, onto a *static* 'reference' image of the environment. Hereafter, it is possible to create Scan-paths (see also: **Scan-paths**) and Heatmaps (see also: **Heatmaps**), or to use Static AOIs for deriving object-related eye tracking statistics (see also: **Static AOIs**).

Examples of 'dynamic' eye-tracking environments are websites (i.e., respondents scroll and navigate through the website pages at their own pace) or eye-tracking data obtained from mobile eye-tracking glasses (i.e., respondents move around freely in the environment).

How to use the Gaze mapping tool

Click [here](#) to download a video

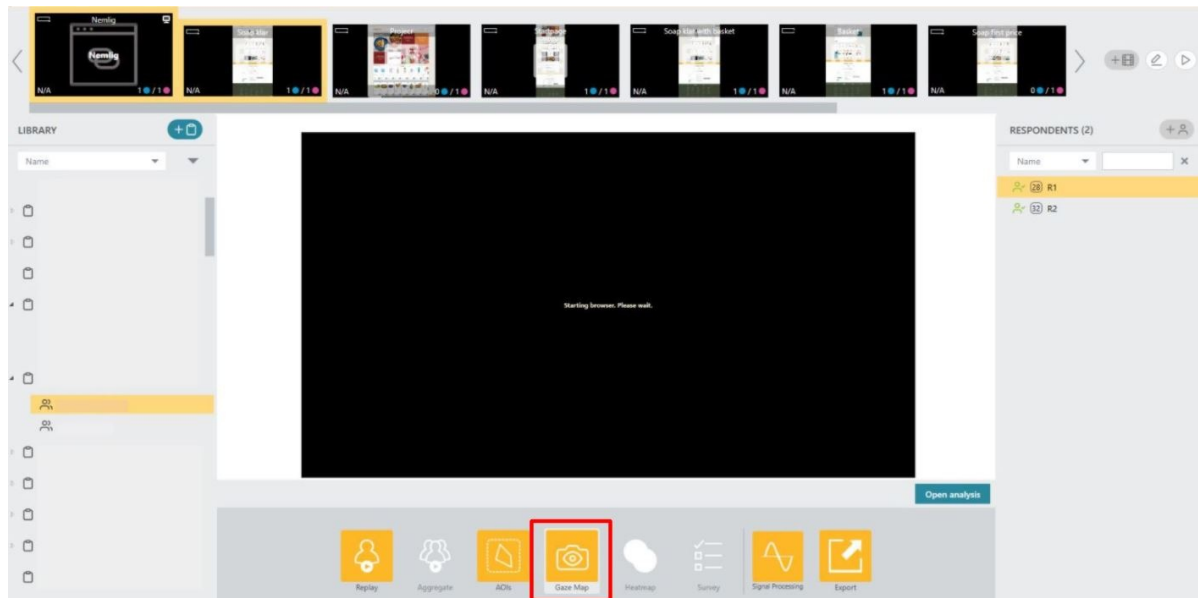
The Gaze mapping tool is available at the 'analysis level' of the iMotions software. This is the part of the software you enter by creating an analysis, or by clicking the name of an existing analysis in the Library.



Enter the Gaze Mapping menu

The Gaze mapping analysis tool is accessible from the lower analysis menu *for dynamic stimulus types*. That includes videos, websites, screen recordings, scene recordings (e.g., mobile eye tracking glasses recordings), as well as 'video scenes' (see also: **Scene editor**).

Gaze mapping is not necessary, and therefore not available, for static stimulus types (e.g., images, survey slides).



How to enter the Gaze Map editor for a dynamic stimulus

To enter the Gaze Map editor *for dynamic stimuli*:

1. Either create a new analysis or select a preexisting analysis for the chosen study
2. Select the dynamic stimuli from the upper menu containing the study's stimuli
3. Click the "Gaze Map" button on the lower menu

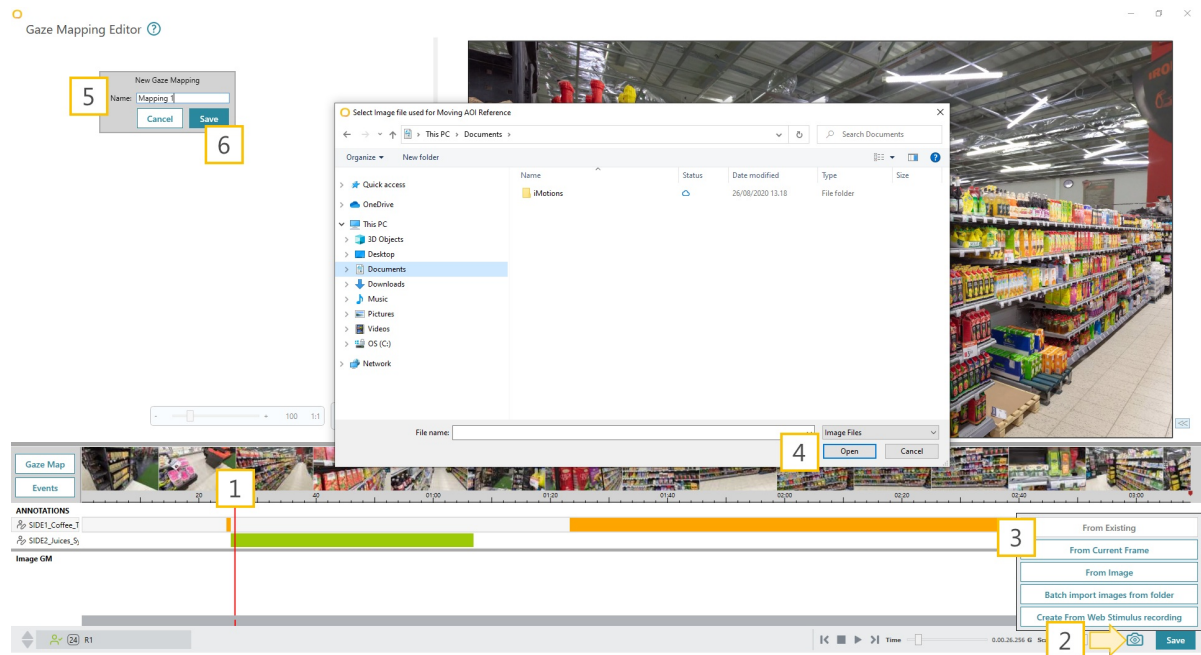
Map *dynamic* eye tracking data onto a *static* 'reference' image

Gaze mapping is about mapping the eye-tracking data obtained in a dynamic environment onto a static 'reference' image. It requires **(1)** choosing a 'reference' image (e.g., a picture of a shelf or a full page screen capture of a website), and **(2)** defining a time interval for gaze mapping.

The 'reference' image will often be identical across multiple respondents. All respondents may have visited the same shelf in a store; all respondents may have visited the same website pages. In contrast, the time interval will often be different across multiple respondents. Some respondents may have passed by the shelf very briefly, while other respondents spent a much longer time looking at the products on the shelf.

That means that, first, you have to define the 'reference' image by choosing one of the following options: **(1)** From existing, **(2)** From current frame, **(3)** From the image, **(4)** Create from web stimulus recording, or **(5)** Batch import images from a folder. Hereafter, you define the time intervals for every respondent individually. Finally, you run the Gaze mapping algorithm.

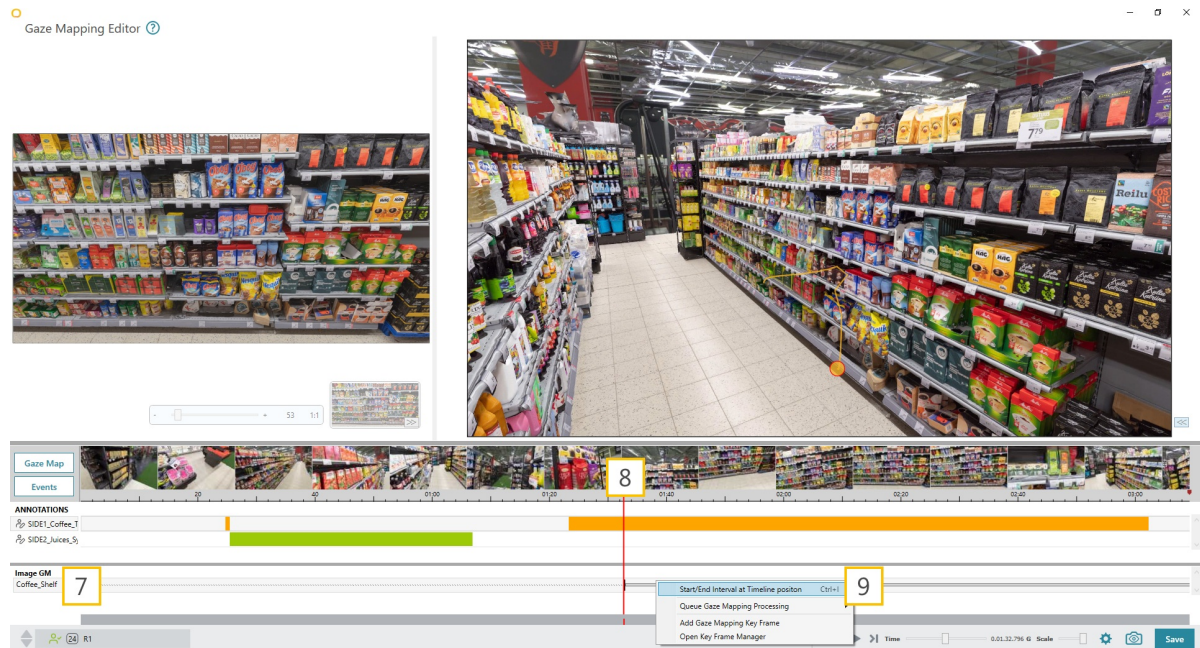
The basic steps are the following.



How to do Gaze Mapping (steps 1-6)

To apply Gaze mapping (yellow path):

1. Set the cursor at the desired start time for the Gaze mapping
2. Click on the Gaze Mapping button at the bottom of the screen
3. Choose one of the following options: From the current frame or the image. The option From current frame will use the current frame as a 'reference' image; the option From image will allow you to upload an image. (The options Batch import images from a folder and Create from web stimulus recording (beta version) are explained later in this article).
4. *If you choose the option From image, browse for the image on your computer.*
5. Then, provide a name for the Gaze mapping image.
6. Click Save. **(The 'reference image is now set')**

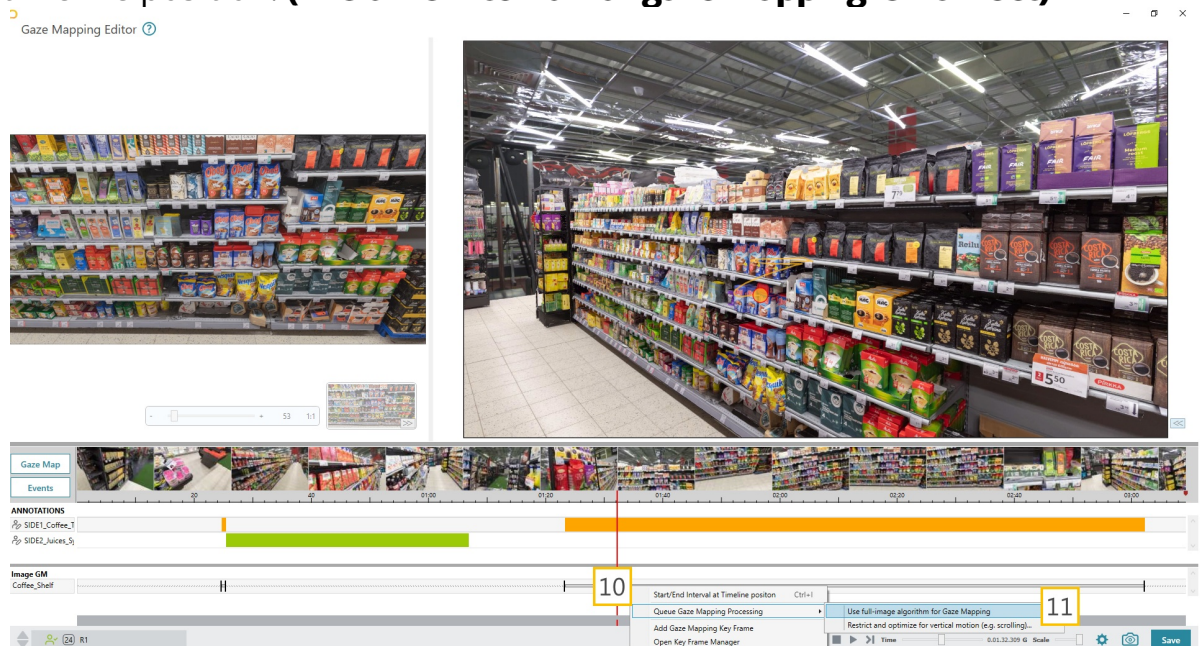


How to do Gaze Mapping (steps 7-9)

7. Note that the image is now added to the timeline, ranging from the cursor time until the end of the timeline.

8. Now, set the cursor at the desired end time for the Gaze mapping. Alternatively, you may choose to use annotation intervals for gaze mapping

9. Open the right-click menu, by right-clicking in the space on the timeline dedicated to the gaze mapping image, and click Start/end interval at the timeline position. **(The time-interval for gaze mapping is now set)**

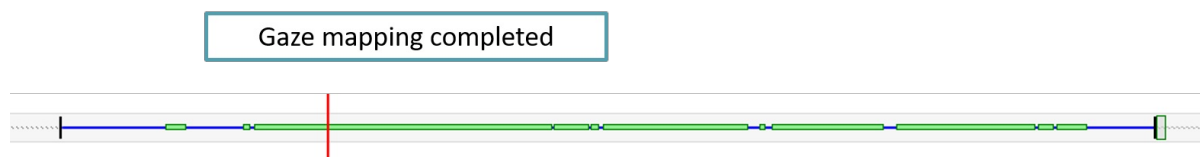


How to do Gaze Mapping (steps 10-11)

10. Finally, re-enter the same right-click menu, and click Queue gaze mapping processing

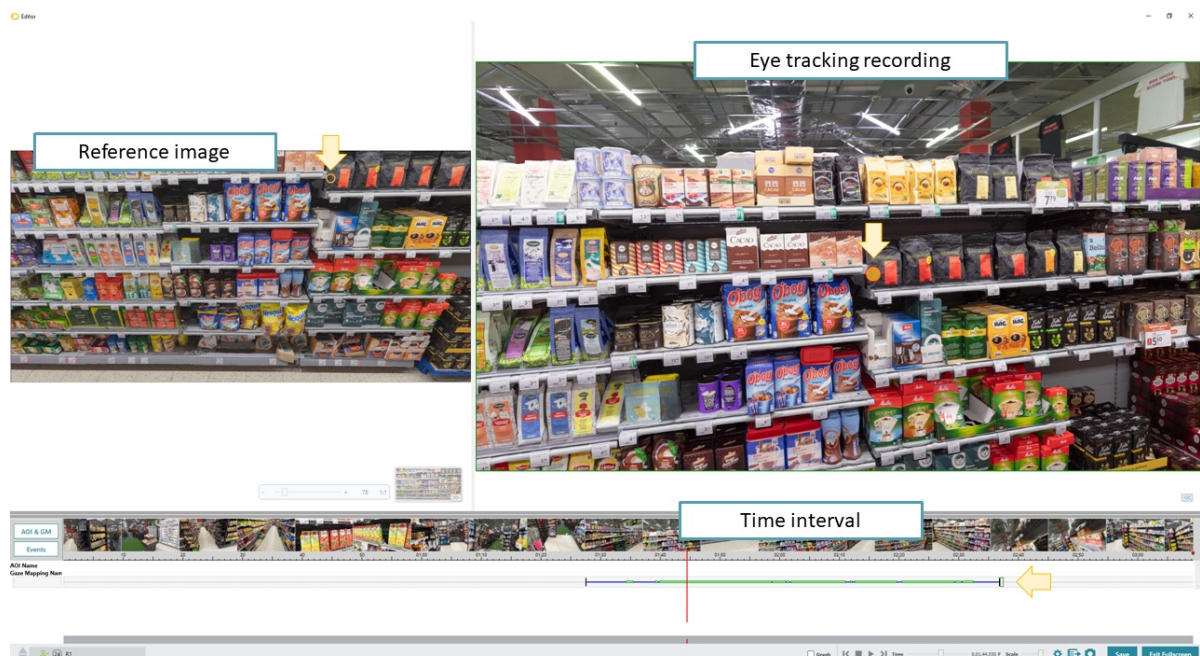
11. Click Use full-image algorithm for gaze mapping. Alternatively, you may choose the option to restrict and optimize for vertical motion (e.g., scrolling). **The gaze mapping algorithm will now start.**

Note that the gaze mapping interval on the timeline colors green if the gaze mapping is successfully completed; blue intervals are periods where the gaze mapping is not successful. However, that does not necessarily mean that the gaze mapping was inaccurate. It could also be that the respondent simply looked outside of the frame corresponding to the 'reference' image.



Gaze mapping completed

You can visually inspect the quality of the mapping by replaying the data. This is done by comparing the gaze points in the original eye-tracking recording (on the right side) and the mapped gaze points in the static 'reference' image (on the left side). If you are not fully satisfied with the result, you may improve the gaze mapping by applying corrections (see below for more information about applying corrections).



How to visually inspect the gaze mapping

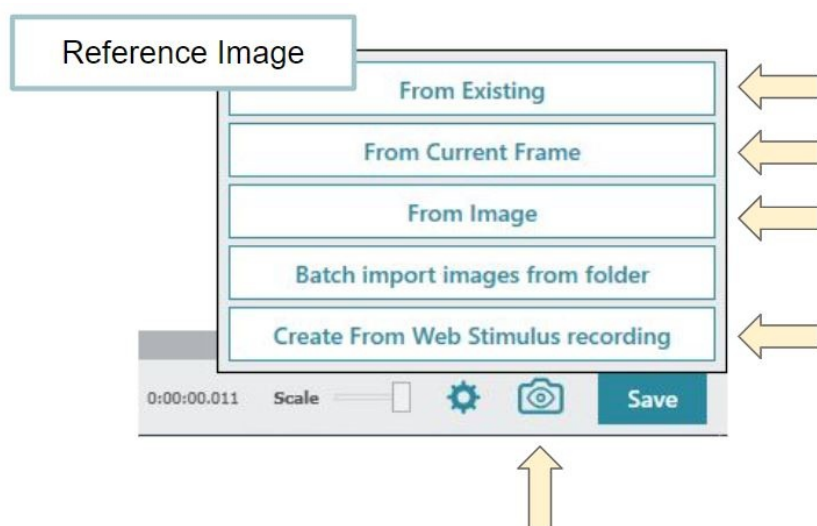
If you are finally satisfied with the result, you have successfully applied gaze mapping *for one respondent*. Now, you should repeat the same steps *for all other respondents*, but choose the option From existing (instead of From current frame or image). This option allows you to apply gaze mapping onto a 'reference' image that is already in use for a different respondent (see below for information about batch processing options). Using the option From existing will also aggregate the individual respondents' data onto a single reference image.

Highlighted features

There are a few options for choosing a 'reference' image for the gaze-mapping: **(1)** From existing, **(2)** From the current frame, **(3)** From the image, and **(4)** Created from web stimulus recording (beta version). The option 'Batch import images from folder' is discussed below.

From existing

The option 'From existing' is only available if you have already uploaded a reference image as part of an earlier Gaze mapping routine. It allows you to continue mapping onto the same reference image that was already defined for previous respondents.



Options for setting a 'reference' image

In other words, you would use the option From existing *for all other respondents*, after having used one of the other options *in a first respondent* (i.e., From current frame, From image, Create from website stimulus recording).

From current frame

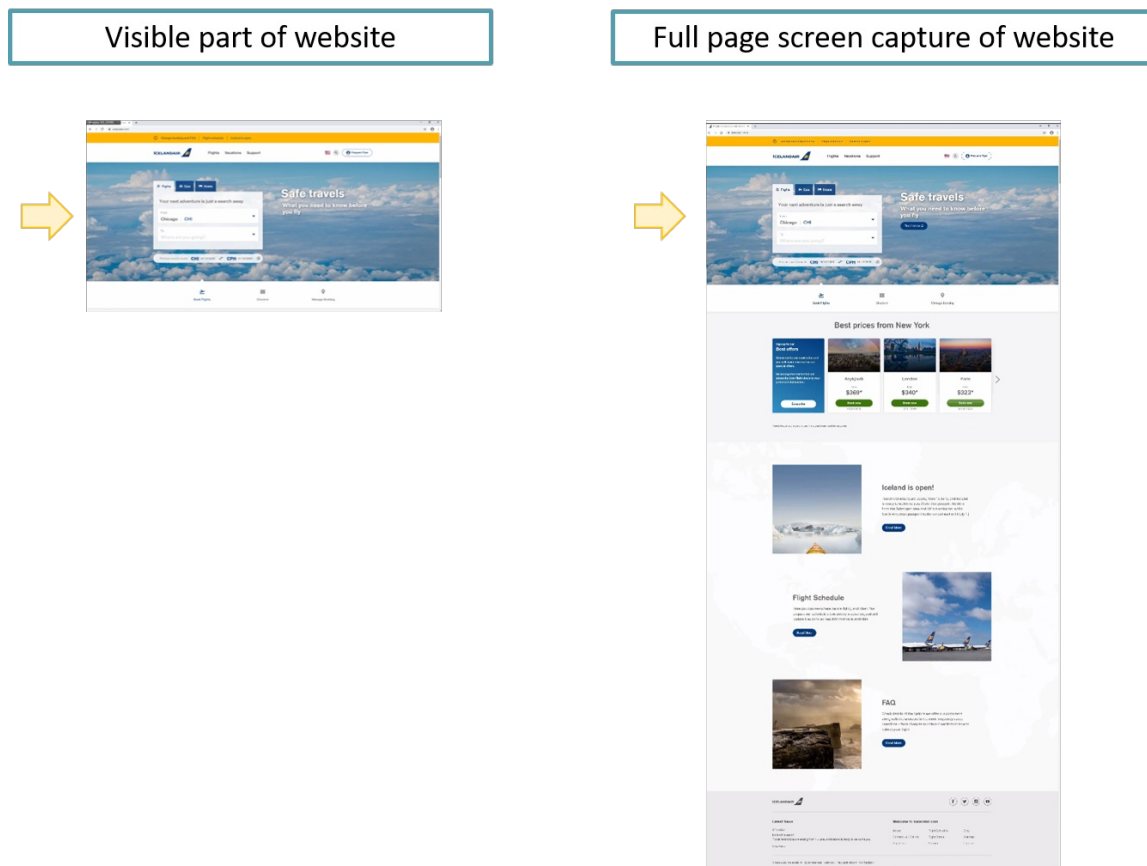
The option 'From current frame' allows you to use one frame of the dynamic stimulus as a reference image.

This functionality is similar to creating 'image scenes' (see also: **Scene editor**).

From image

The option 'From image' allows you to upload a reference image from your computer.

This functionality may also be handy for websites, as you could upload a Full-page screen capture of the website. This means that you collect the page screen capture images *using an external tool outside of iMotions*, and hereafter, you upload these screen captures as images for Gaze mapping.

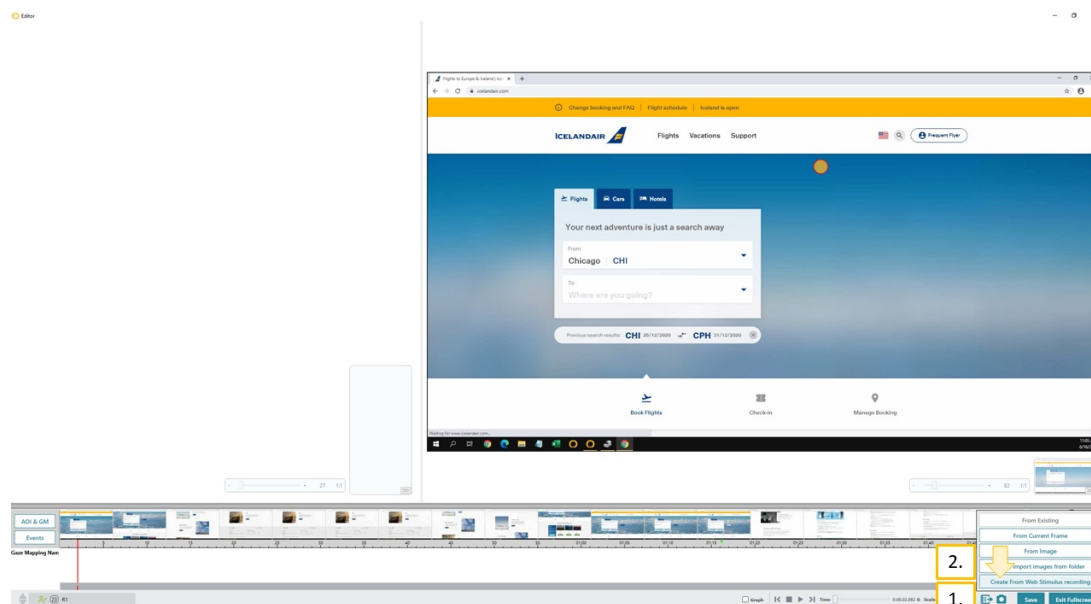


Example of a Full page screen capture of a website

Create from web stimulus recording

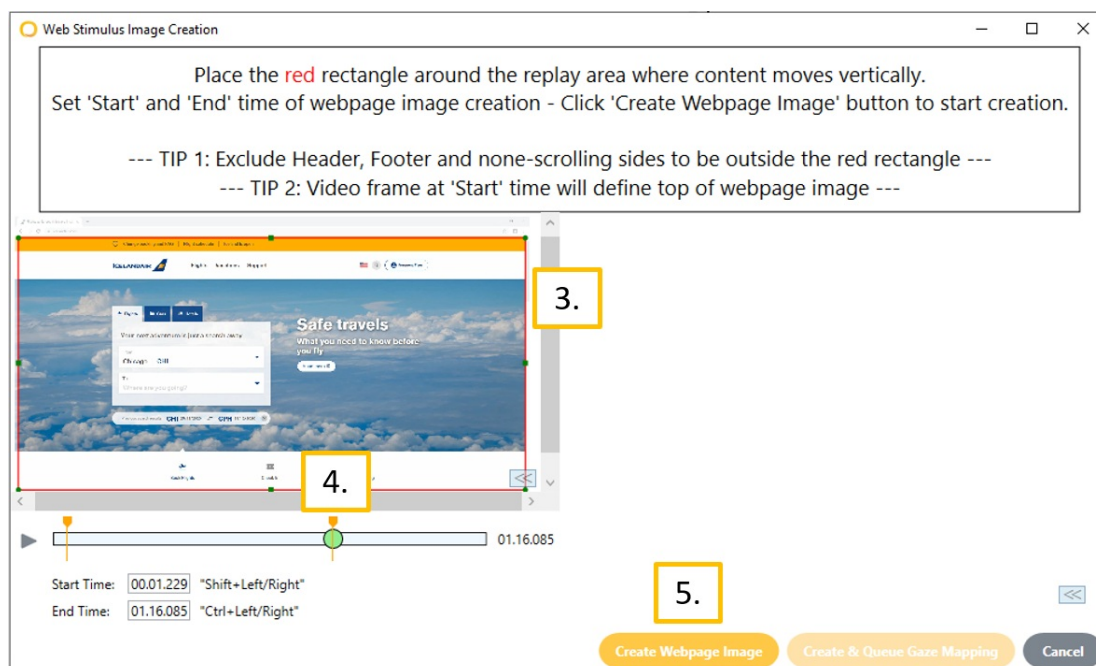
Alternatively, for websites, you may choose the option Create from web stimulus recording. This option can be used to create a reference image of the website page *from within the iMotions software* (instead of using an external tool for obtaining a Full page screen capture). Hereafter, the reference image is immediately also used for gaze mapping.

To gaze map using the option 'Create from website stimulus recording' (yellow path):



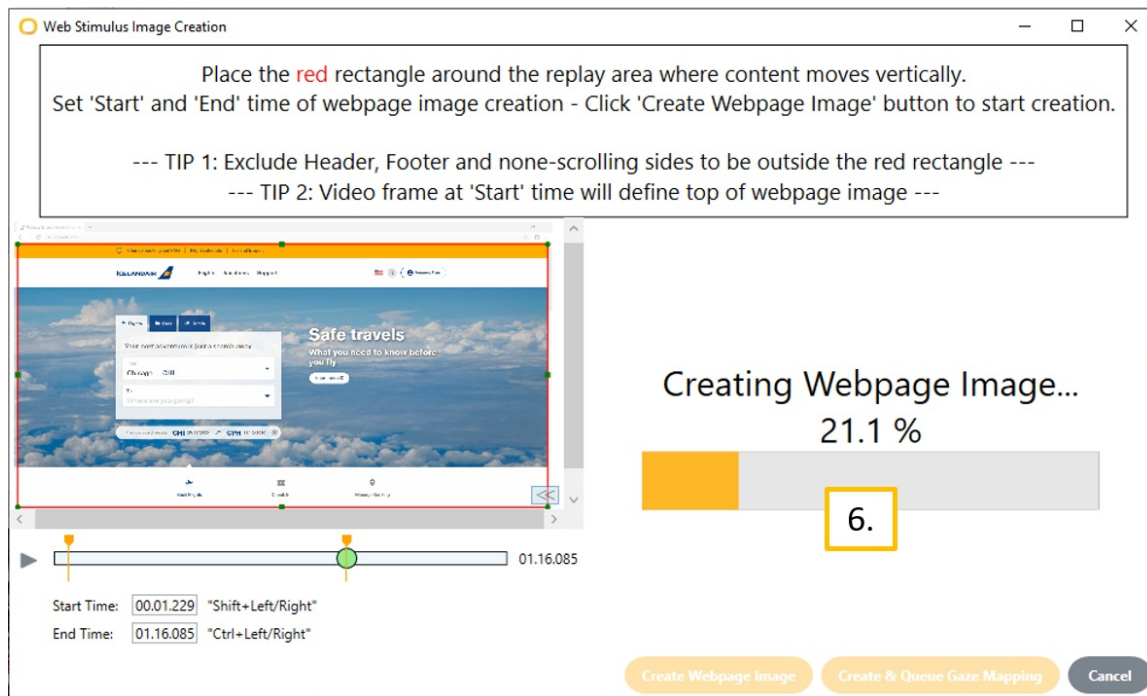
How to create from web stimulus recording (steps 1-2)

1. Click on the Gaze Mapping button in the bottom of the screen
2. Choose the option: Create from web stimulus recording



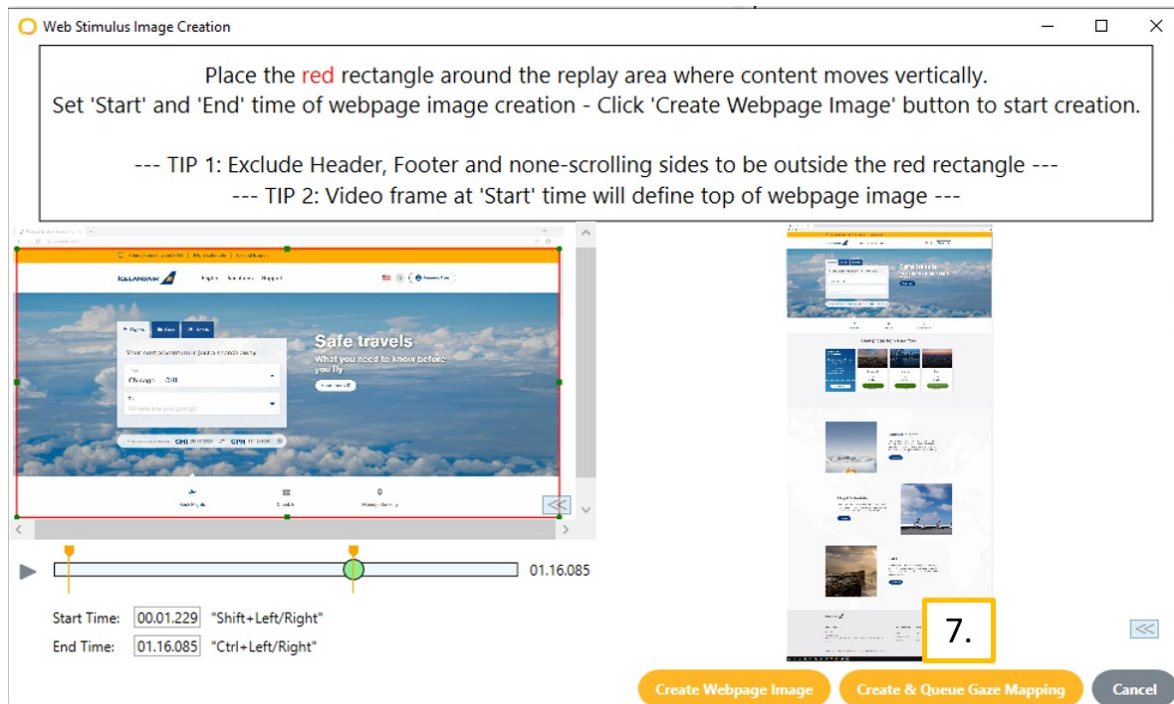
How to create from web stimulus recording (steps 3-5)

3. Place the red rectangle over the moving part of the website
4. Define the time interval by dragging the borders of the interval
5. Click 'Create webpage image'



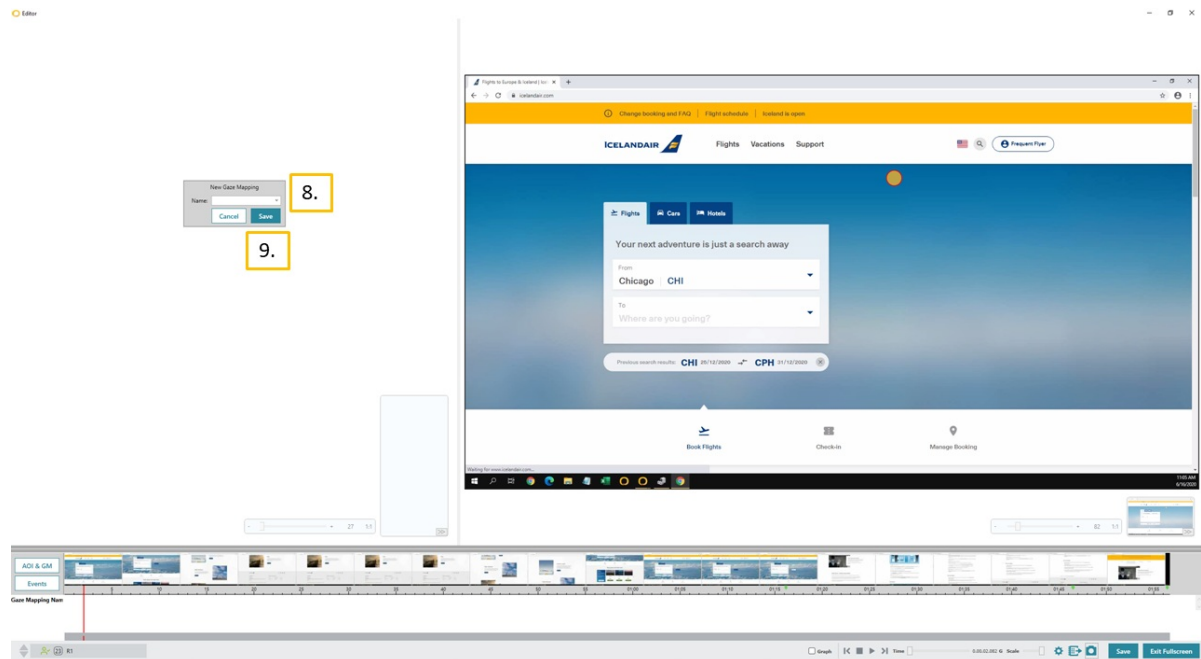
How to create from web stimulus recording (step 6)

6. Wait for the webpage image to be created



How to create from web stimulus recording (step 7)

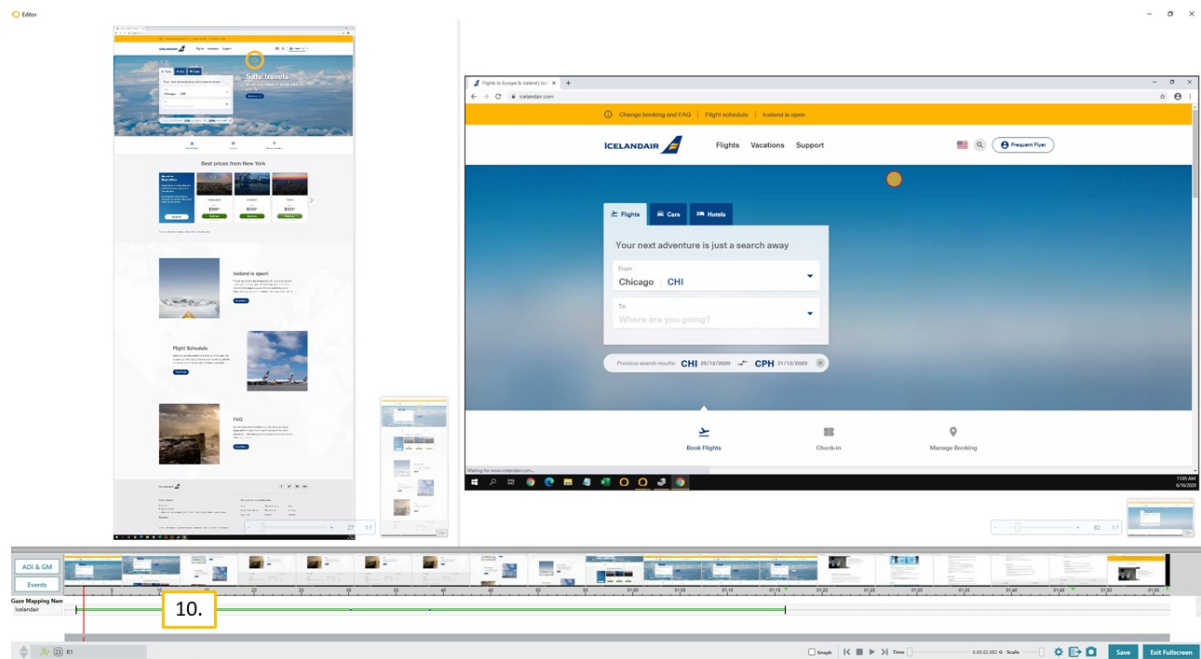
7. Click 'Create and queue gaze mapping'



How to create from web stimulus recording (steps 8-9)

8. Choose a name

9. Click Save



How to create from web stimulus recording (step 10)

10. Wait until the gaze mapping interval colors green.

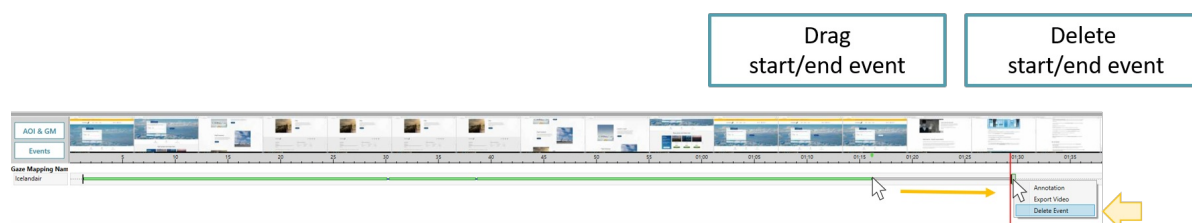
There are some limitations to this tool, for example, the algorithm only considers the information that *occurred on screen*. Hence, if a respondent did not scroll to the bottom of the page, this part of the website will not be part of the reference image. Therefore, it is good best practice to find a suitable respondent from whose recording you generate the reference image, and this respondent shall have scrolled down the farthest. Also, if the respondents scrolled very quickly, the algorithm may not be able to generate a continuous website capture. This may also happen if the page was “interrupted” by a video.

A benefit of creating the reference image directly from the web stimulus recording is that it allows you to generate static gaze plots on websites that were unique for the current respondent and therefore impossible for you as a researcher to create before or after the session. An individual's newsfeed, for example, is constantly changing, but this tool allows you to create a reference image from the individual recording. Another benefit of this tool is also that it can handle floating header and footer page content better than third-party browser plug-ins, as one can simply not include the static header and footer when selecting the frame of the website to include.

Often a better alternative for website gaze mapping is to create a 'web scene' (see also: [Scene editor](#)); this is the third available option for gaze mapping on websites.

Change start/end time

For defining the start and end time of the gaze mapping interval, you would place the cursor at the desired times on the timeline and use the start/end button in the right-click menu. The start time would typically be defined *before* adding a 'reference' image, and the end time *after* adding the 'reference' image. Alternatively, you could also change the start and end of the time interval by dragging the borders of the gaze mapping interval, or you could delete an event.



How to change start/end times

To drag the start and end time of the gaze mapping interval:

1. Hover over the start/end time, and click on the key frame (or 'event')

2. Hold and drag the event

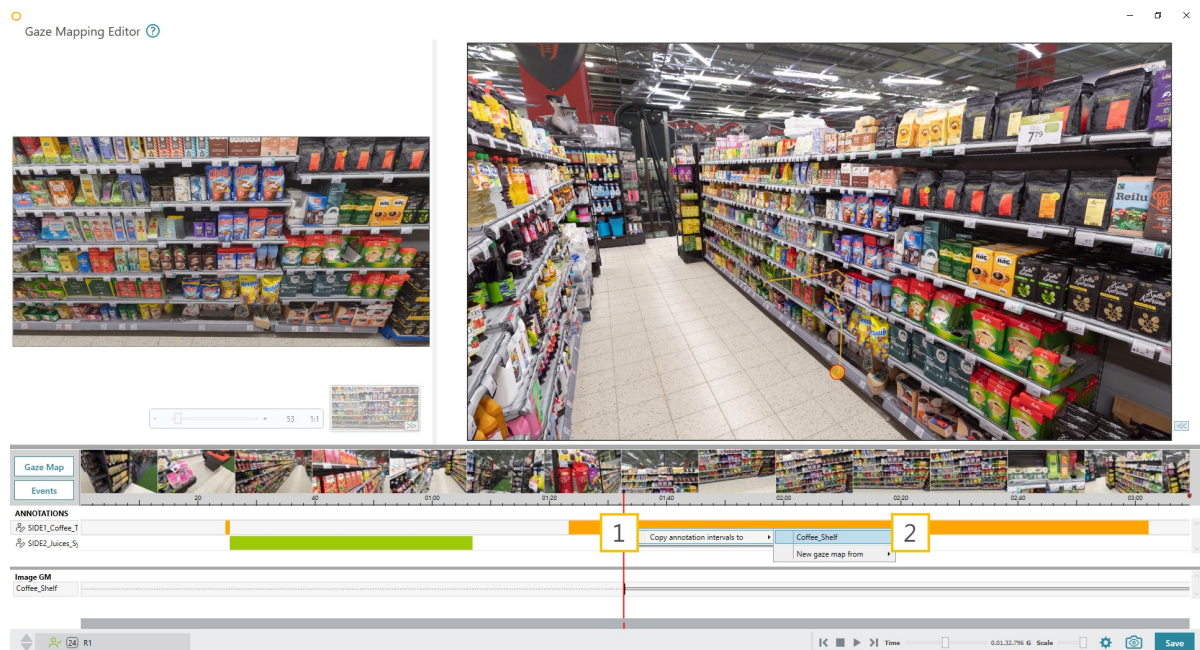
To delete a start or end event:

1. Hover over the start/end time, and right-click on the key frame (or 'event')
2. Click 'Delete event'

Note that you may now have to restart the gaze mapping algorithm for the part that has not been gaze mapped yet.

Use annotation intervals for gaze mapping

If you have already created annotations in the Replay or Aggregate editors, your annotations will be shown in the Gaze Mapping editor. (Note that you cannot create, edit, or delete annotations from within the Gaze Mapping editor, only from the Replay or Aggregate editors) This may save you time, as if you have already annotated the timeline, you can quickly copy over those specific timeline intervals without having to set it manually for your Gaze Mappings.



How to use annotation intervals for a gaze mapping

You can use any annotation to set the intervals of a gaze mapping. To do this:

1. Hover over the annotation that you want to use for your gaze mapping intervals, and right-click on that annotation

2. Select 'Copy annotation intervals to' and then the name of the gaze mapping you would like to use. **If you have not yet created a gaze mapping, you also have the option to create one directly from the annotation, by selecting the 'New gaze map from' item, and then selecting the type of reference image you would like to use.*

Note that you may now have to restart the gaze mapping algorithm for the part that has not been gaze-mapped yet.

Delete a gaze mapping

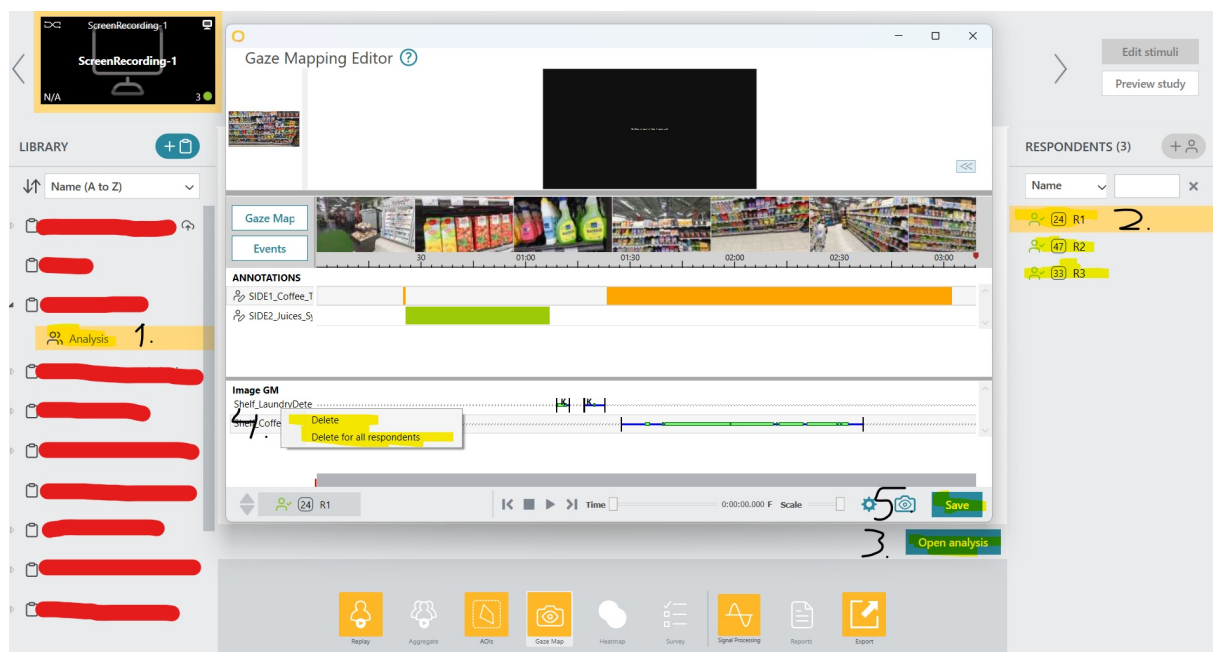
It is finally also handy to know how to delete a gaze mapping. You may either want to delete the gaze mapping *for one respondent*, or *for all respondents*.

Deleting a gaze mapping for one respondent is done by removing it from the timeline (for a given respondent).

To delete a gaze mapping *for one respondent*:

1. Click on the Analysis for the desired study
2. Choose the desired respondent on the right
3. Click on "Open Analysis" on the right-hand sight
4. Right-click on the image that you want to delete.

In contrast, deleting the gaze mapping *for all respondents* is done by deleting the reference image from the Gaze Mapping Editor.



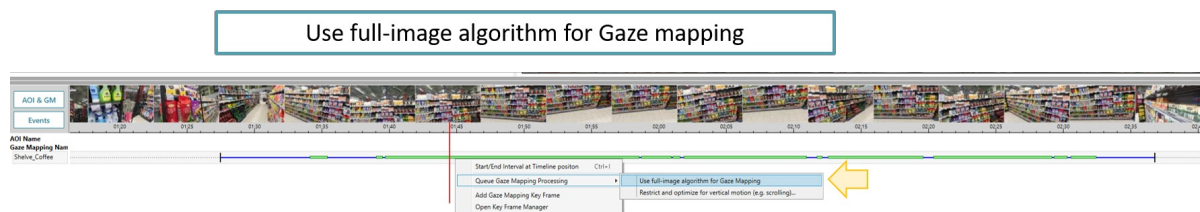
How to delete a gaze mapping for one or all respondent

Algorithms

There are two algorithms for doing the gaze mapping: **(1)** a standard algorithm, and **(2)** an algorithm optimized for 'vertical motion' (scrolling).

Use full-image algorithm for gaze mapping

The standard algorithm for Gaze mapping is this Full-image algorithm for gaze mapping. This algorithm is essentially an object recognition algorithm that searches for the visual features of the 'reference' image in the original recording. If the algorithm is able to localize the 'reference' image in the eye tracking recording, subsequently, it is capable of mapping the original gaze-points onto their corresponding location in the 'reference' image.

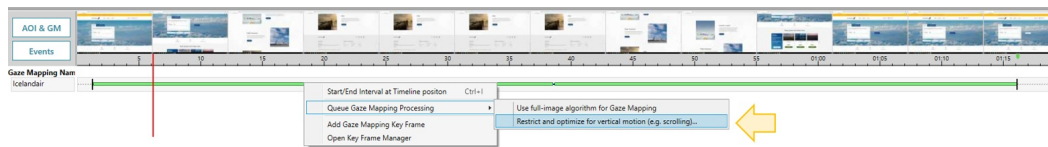
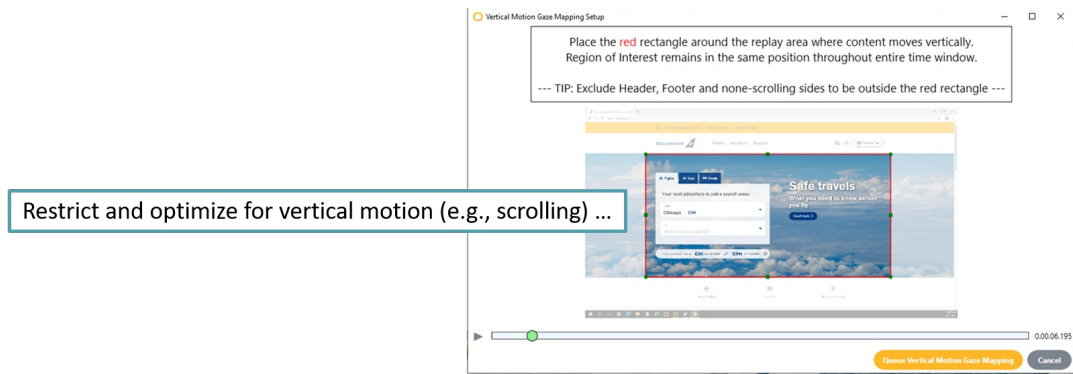


Standard algorithm

Note that there are a few reasons why the standard algorithm for Gaze mapping would not be capable of mapping the eye tracking data: **(1)** the 'reference' image is not present in the video recording made by mobile eye tracking glasses because the respondent looked away from the frame, **(2)** the 'reference' image is present in the video recording, but the respondent does not look inside the frame of the 'reference' image (i.e., the gaze-point falls outside of the 'reference' image), **(3)** the 'reference' image is not similar enough to the original recording environment, or **(4)** the quality of the 'reference' image and video recording are not similar enough (e.g. blurred video-frames due to fast movements).

Restrict and optimize for vertical motion (e.g., scrolling, ...)

The alternative algorithm for Gaze mapping is the Restrict and optimize for vertical motion (e.g., scrolling, ...) algorithm. This algorithm is mostly relevant for websites; in particular, websites with side banners that do not move together with the rest of the website when scrolling down ('vertical motion').



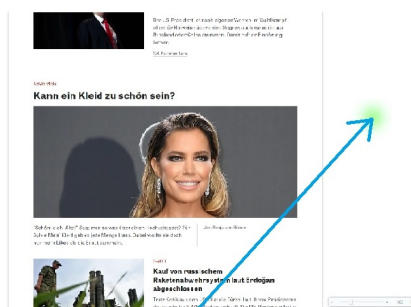
Algorithm optimized for websites (e.g., scrolling)

In such a case, the Restrict and Optimize for vertical motion (e.g., scrolling, ...) option allows you to define what part of the website is dynamic (i.e., the information changes when scrolling down) and what part of the website is static (i.e., the information does not change when scrolling down). This is done by placing a square shape over the *dynamic* part of the website; this is typically the central part of the website (excluding menus at the top, side banners, cookies at the bottom of the screen, etc.). The algorithm then automatically applies a Gaze mapping routine in two waves; a first Gaze mapping routine for the dynamic part of the website (assuming vertical motion), and a second Gaze mapping routine for the static part of the website (assuming the absence of vertical motion).

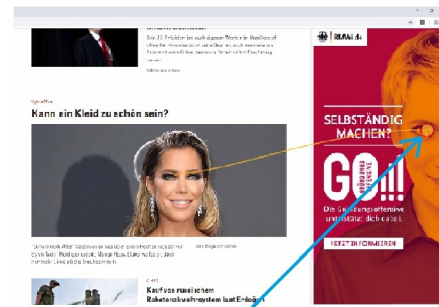
What is the benefit of the algorithm optimized for vertical motion?

When a respondent scrolled on a website (or screen recording), then it often happens that parts of the website remain where they are (for example an ad banner on the side, or a menu at the top) whereas other parts get moved by the scrolling.

When the algorithm then tries to map the gaze data from the recording onto the reference image, this can lead to faulty mappings because objects that were next to each other in the recording are not next to each other in the reference image.



Gaze point gets mapped in relation to the picture that was next to it, but does not get remapped onto the ad. So it is difficult to analyse where on the ad the respondent was focusing on if the ad changed its position relative to the other content.



In the replay of the recording, one can see that the respondent looked at the ad.

Through running two separate mappings, however, this can be corrected.



Remapped gaze on reference image



Original gaze replay in recording

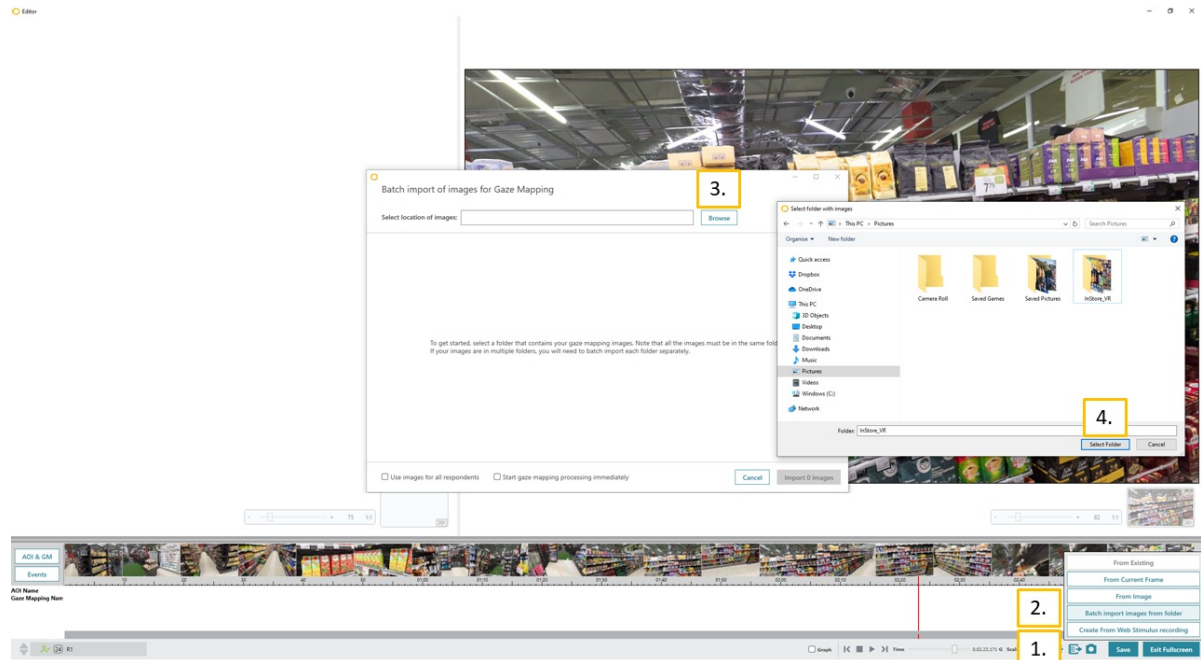
Batch processing options

Defining the time intervals for gaze mapping is mostly manual work, but there are options for batch importing 'reference' images from a folder, and for applying the gaze mapping algorithms as a batch processing job. The latter can be done on your local computer, or using the cloud (requires a special license).

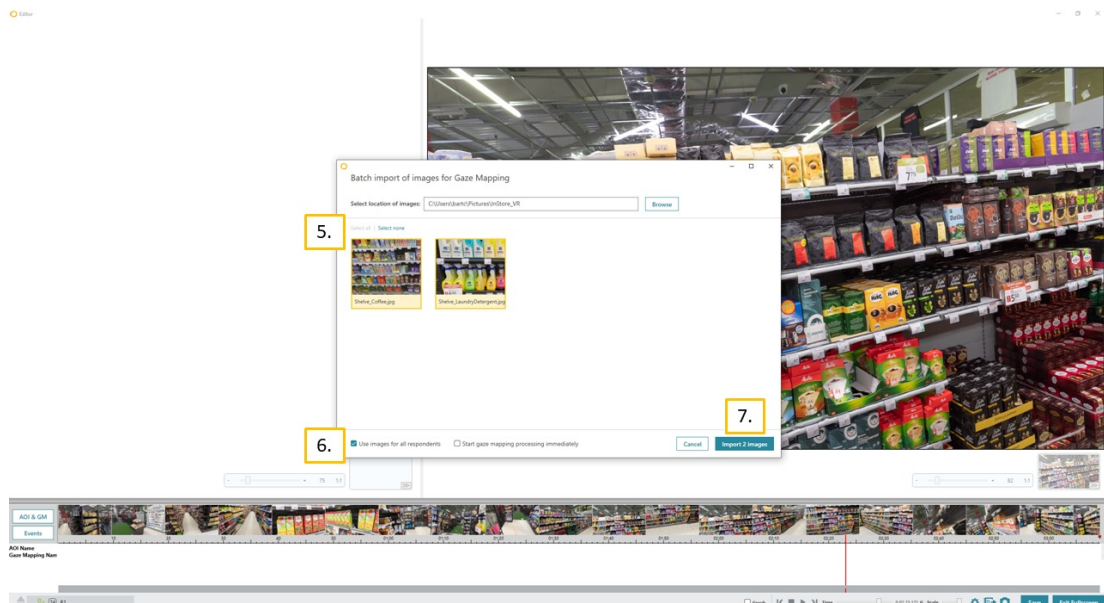
Batch import images from the folder

If you anticipate running a Gaze mapping on multiple reference images ('From image'), you could choose to load the reference images all at once using the option Batch import images from a folder. The option is also handy if you upload only one reference image but it applies to all respondents in the analysis; by batch importing the reference image, it can immediately be loaded for all respondents, rather than using the 'From existing' option in every respondent individually.

To batch import reference images (yellow path):

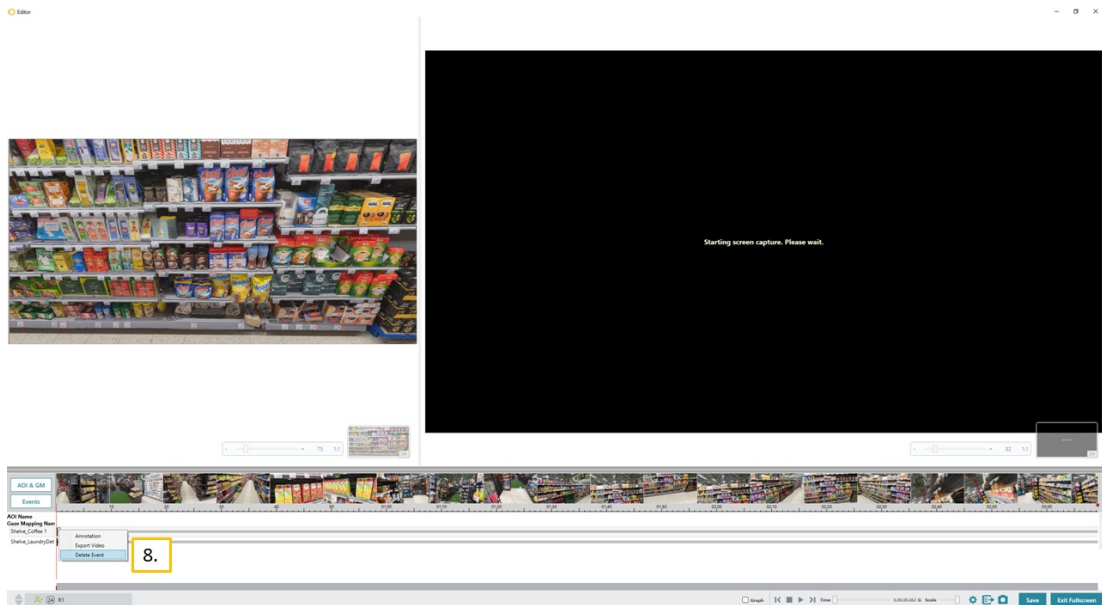


1. Click on the Gaze Mapping button at the bottom of the screen
2. Choose the option: Batch import images from a folder.
3. Click Browse
4. Browse for the folder on your computer that contains the reference images.

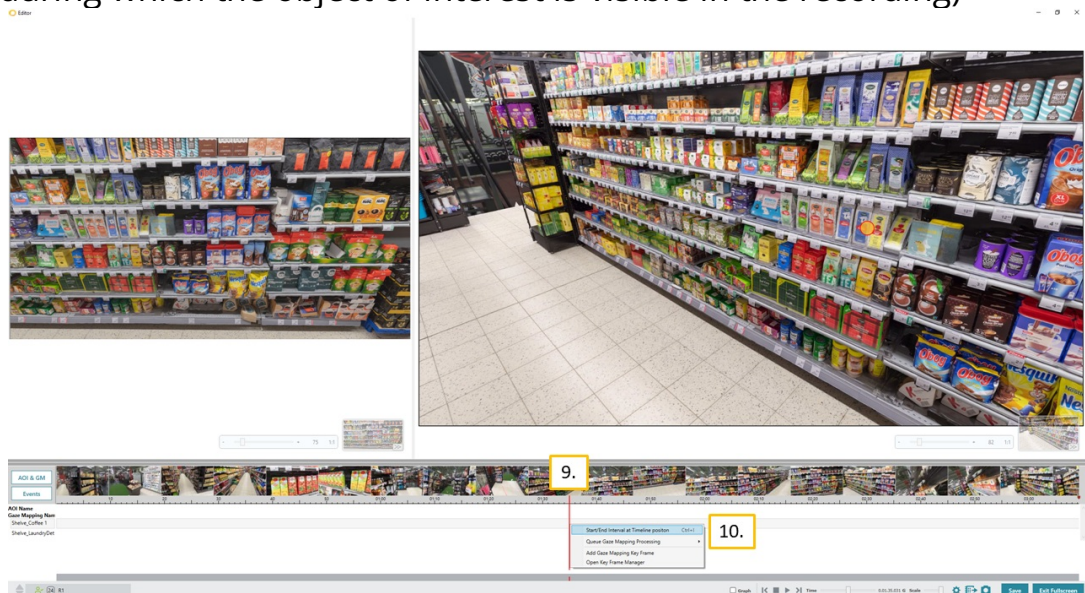


5. Use the Select all | Select none button, select a single image, or hold down 'CTRL' and select multiple images.
6. *If relevant*, tick the option Use images for all respondents.

7. If relevant, tick the option Start gaze mapping processing immediately. We recommend this option only if you have short stimuli or if the content of the reference images is visible throughout the majority of the recordings. In most cases, we would recommend to leave this deselected and first set the timelines (see step 9 and following).
8. Then, click Import images.

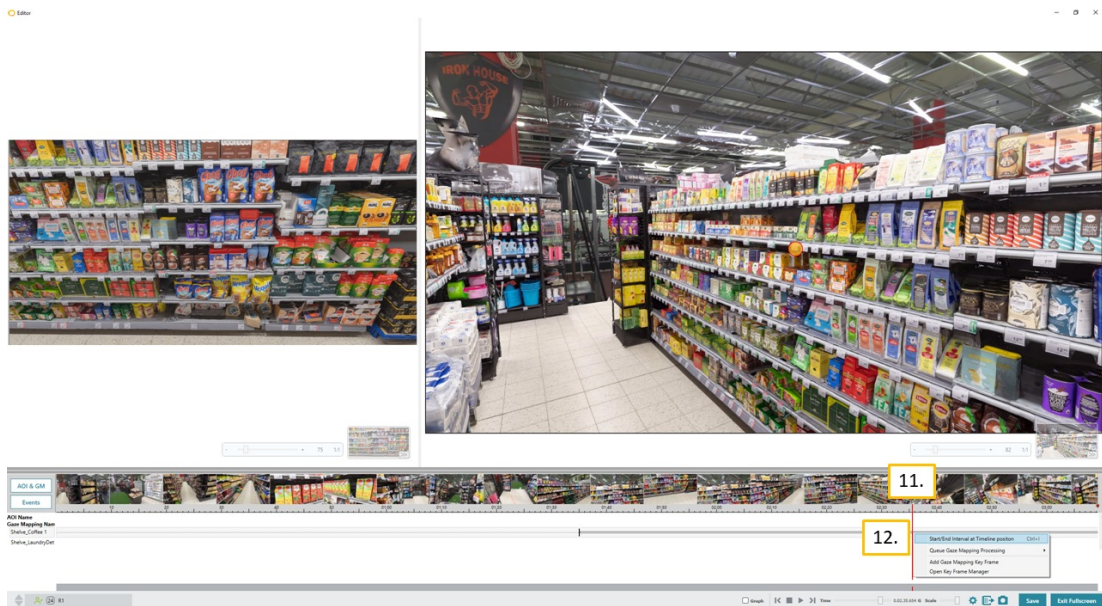


9. *If the gaze mapping should not start from the beginning of the recording, delete the event at the start of the recording. (We recommend only setting the gaze mapping timeline for the period during which the object of interest is visible in the recording)*



10. Set the cursor at the desired start time for the Gaze mapping

11. Right-click in the space on the timeline dedicated to the gaze mapping image, and click Start/end interval at timeline position



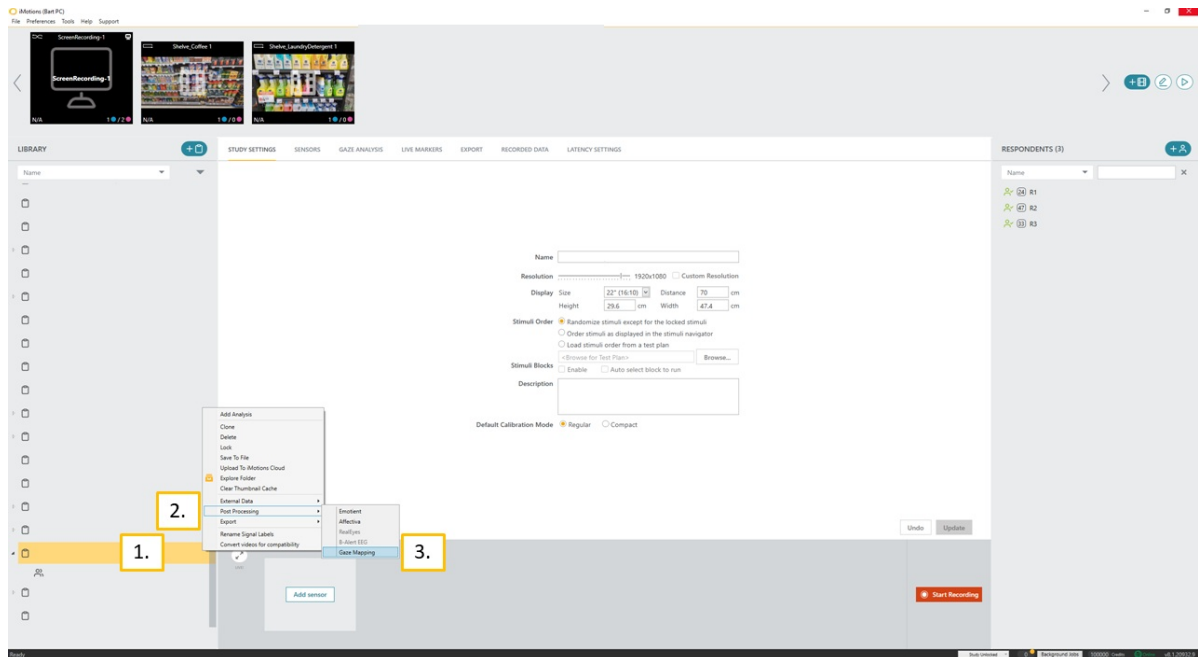
12. Set the cursor at the desired end time for the Gaze mapping
13. Right-click again in the space on the timeline dedicated to the gaze mapping image, and click Start/end interval at the timeline position

You should now correct the start/end times of the Gaze mappings for all respondents individually. Hereafter, you may choose to run the Gaze mappings for all respondents individually, or to Batch process all Gaze mappings at once.

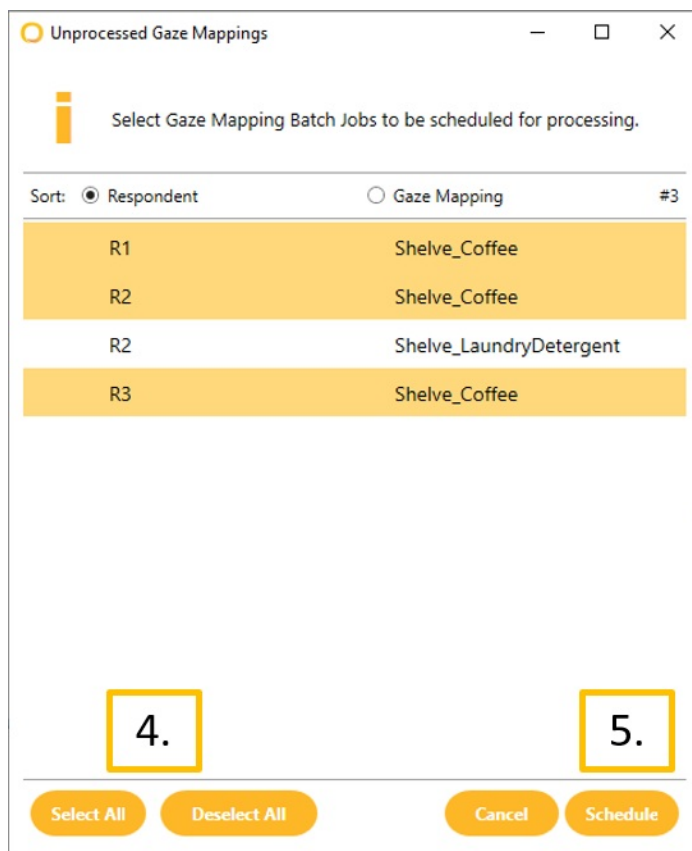
Batch processing

Suppose the 'reference' images are loaded and the time intervals are set for all respondents. In that case, you may choose to Batch process all the Gaze mappings (instead of starting the Gaze mapping for all respondents individually).

To batch process the gaze mappings (yellow path):



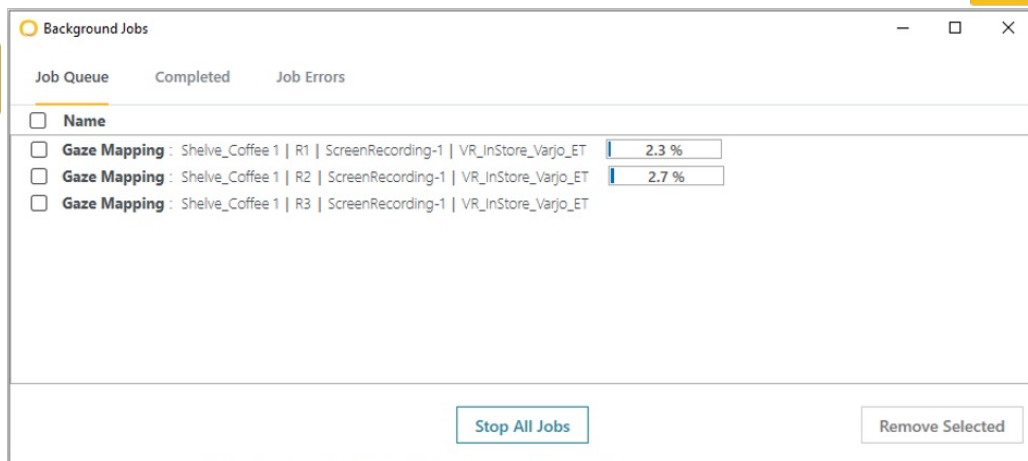
1. Exit the Gaze Map editor, and right-click on the name of the study in the Library.
2. Click Post-processing
3. Click Gaze mapping



4. Use the options Select all | Deselect all, or select the relevant gaze mappings individually.'

5. Click Schedule.

6.

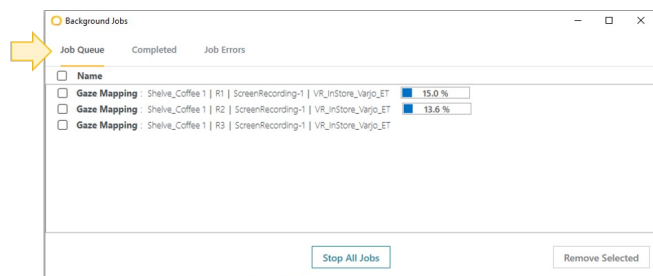


6. Verify whether the Batch processing starts successfully, by checking the job queue.

7. Exit the window

Gaze mapping is a background job, which means that you can continue using the iMotions software in the meantime. However, you should **not** close down the iMotions software; that will discontinue the process.

Check the background jobs



Study Unlocked 3 Background Jobs 100000 Credits v8.1.20932.9

How to check the progress

To check the ongoing background jobs:

1. Click the Background Jobs button in the bottom of the iMotions software
2. Check the progress in the job queue.

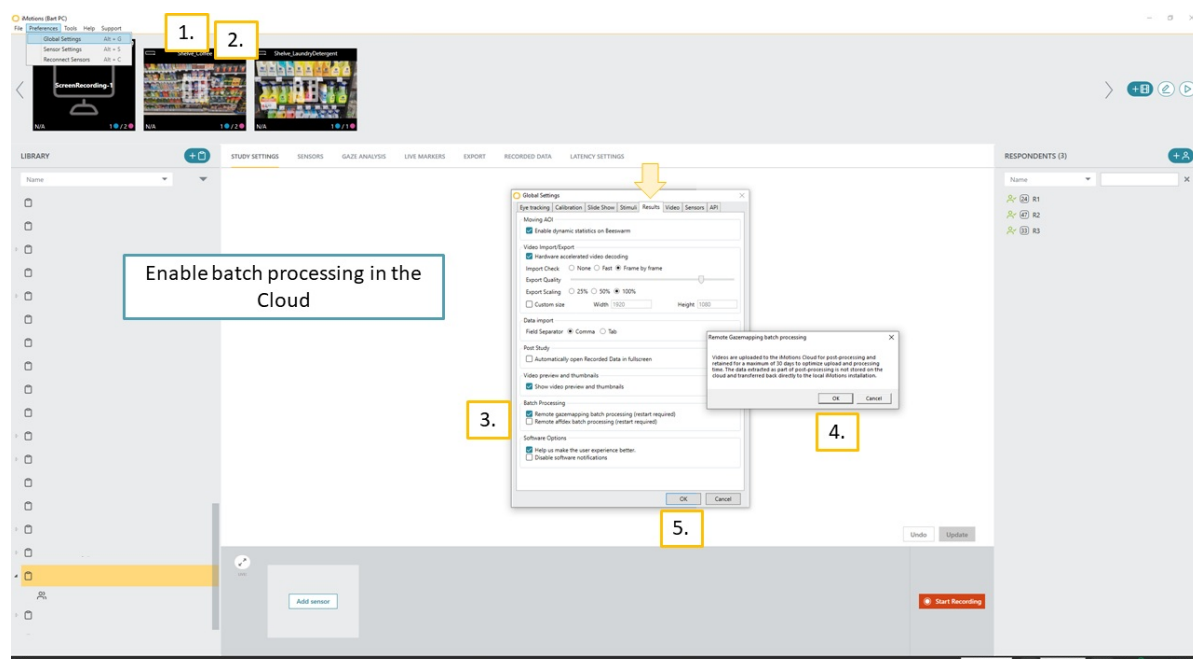
You can reorder the background jobs manually and thereby change the order in which Gaze Maps are being processed. If you drag a new job to the top, this will replace and pause the processing of the current job. The progress will NOT be lost.

Batch processing using the cloud (requires license)

Instead of doing the processing on your computer, you may also want to use the cloud for batch processing. However, that requires a special license.

Batch processing in the cloud is done in the same way as batch processing on your computer, except that, in the background, your data is first uploaded to the cloud, then processed in the cloud, and eventually, downloaded back to your computer. Note that the uploaded video materials are retained in the cloud for maximally 30 days (this is to save user upload time if multiple gaze mappings are made on the same video); the gaze mappings themselves are never stored in the cloud and returned immediately to your server.

If you have the license, you may choose to use the cloud for batch processing by enabling the option in Global Settings. This is done in the following way.



How to enable batch processing using the cloud (requires license)

To enable batch processing using the cloud (yellow path):

1. Go to Global settings, by clicking 'Preferences'
2. Then, click 'Global settings'

3. Navigate to the 'Results' page, and tick the option 'Remote gaze mapping batch processing (restart required)'
4. Accept the warning message about data usage, by clicking 'Ok'
5. Click 'OK' to leave Global settings

Hereafter, you should restart the iMotions software.

Now, the Batch processing of the gaze mappings can be started in the same way as regular Batch processing.

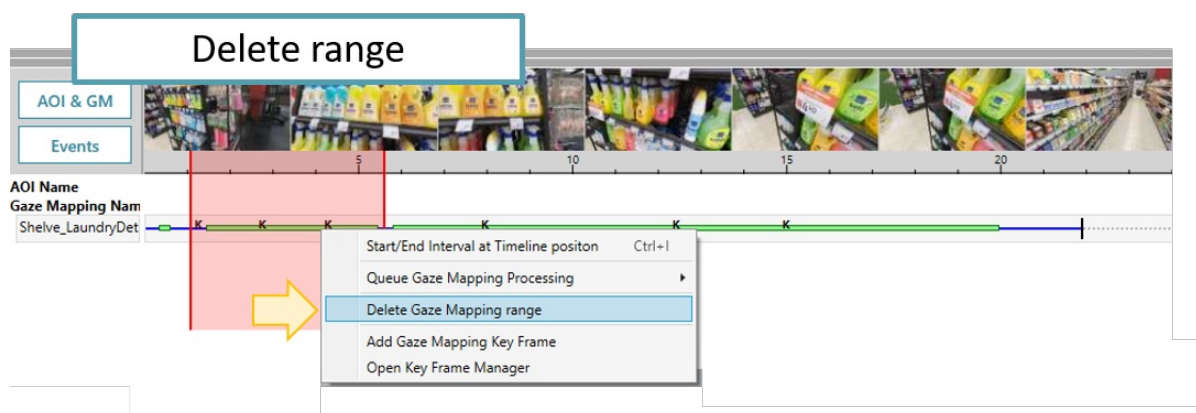
Applying corrections

Finally, we note that the mapping of the eye tracking data onto a 'reference' image is a challenging task, and the mapping will not always be successful. Sometimes the mapping will be inaccurate (i.e., the 'reference' image was identified successfully in the eye tracking recording but the mapping of the gaze point is incorrect); sometimes the algorithm will simply not be able to identify the 'reference' image.

Two types of corrections can be applied in those cases: **(1)** adding keyframes, and, if that does not help, **(2)** manual corrections. However, first, you may want to delete the parts of the gaze mapping that are done incorrectly.

Delete a gaze mapping range

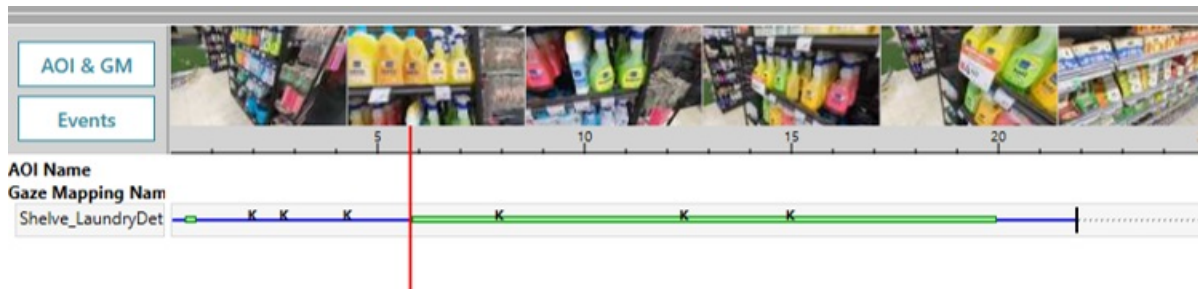
If the mapping is inaccurate for a longer period, you could delete this part of the gaze mapping in the following way.



How to delete a gaze mapping range

To delete a part of the gaze mapping:

1. Click in the area on the timeline that corresponds to the Gaze mapping you are currently working on. Make sure that the red horizontal line is placed at the start of the part you want to delete.
2. Hold down 'SHIFT'
3. Now, click again in the same zone, but at the end of the part you want to delete. Note that the range is now colored red.
4. Finally, right-click in the same zone, and click Delete gaze mapping range.



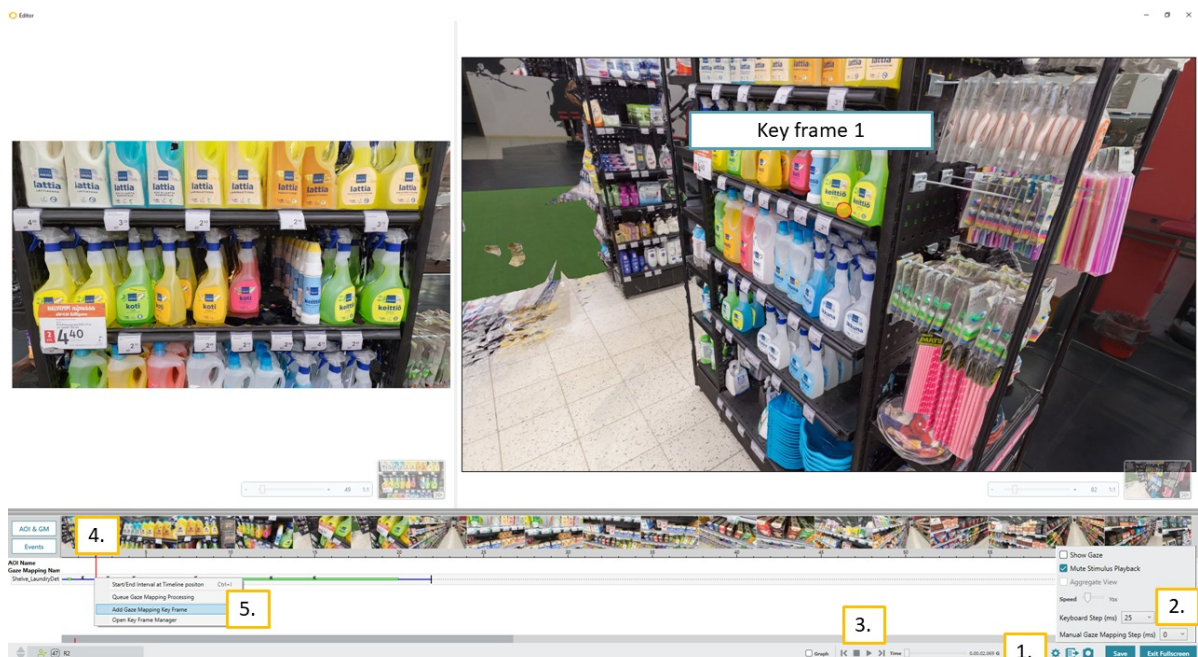
The result of deleting a range

Note that the deleted part of the gaze mapping is now colored blue.

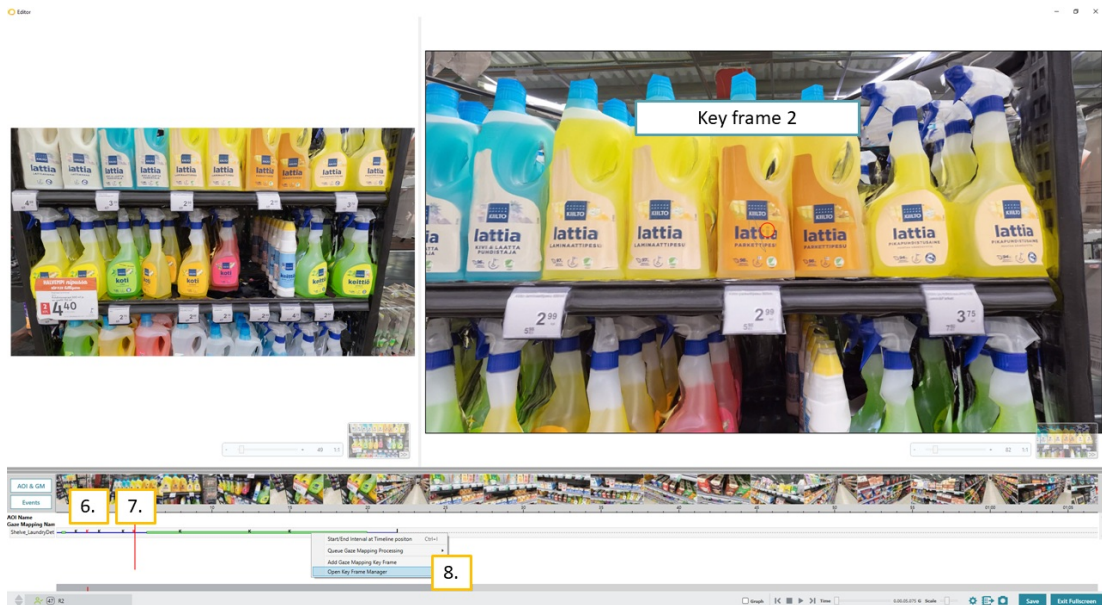
Keyframes

Hereafter, you could place a few keyframes and repeat the gaze mapping routine. Keyframes help the Gaze mapping algorithm to identify the 'reference' image in the eye tracking recording. This is done by selecting a few frames in the eye tracking recording ('key frames'), and then, manually localizing the features of the 'reference' image in the recording.

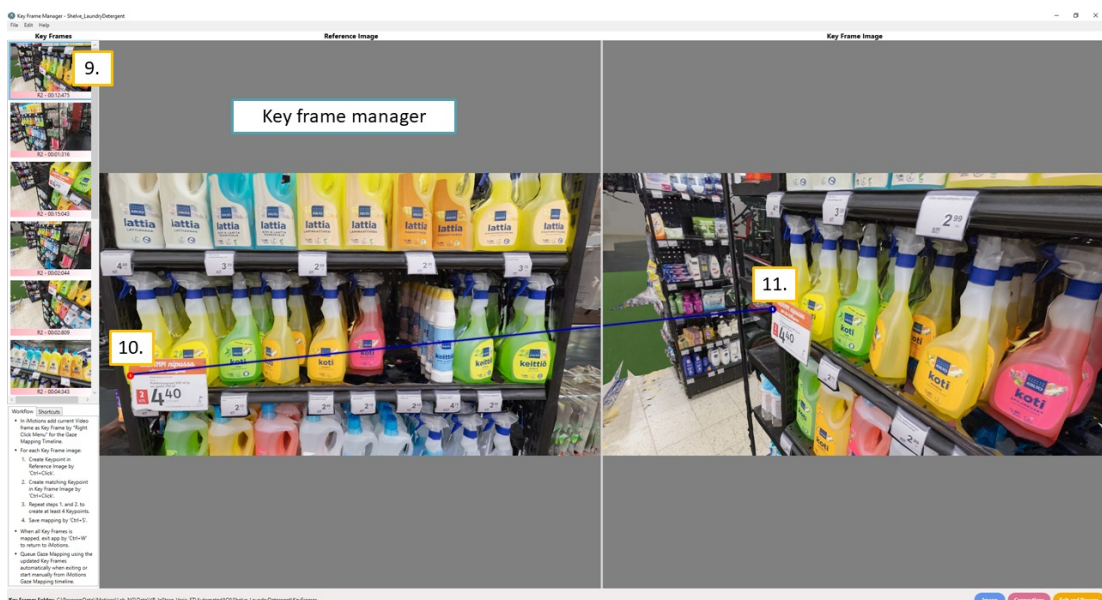
To add keyframes (yellow path):



1. Click the Settings button at the bottom of the screen.
2. Navigate frame-by-frame through the recording to identify frames that could be used as keyframes.
3. If you find a suitable frame, right-click in the area on the timeline that corresponds to the Gaze mapping you are currently correcting.
4. In the right-click menu, click Add gaze mapping keyframe.

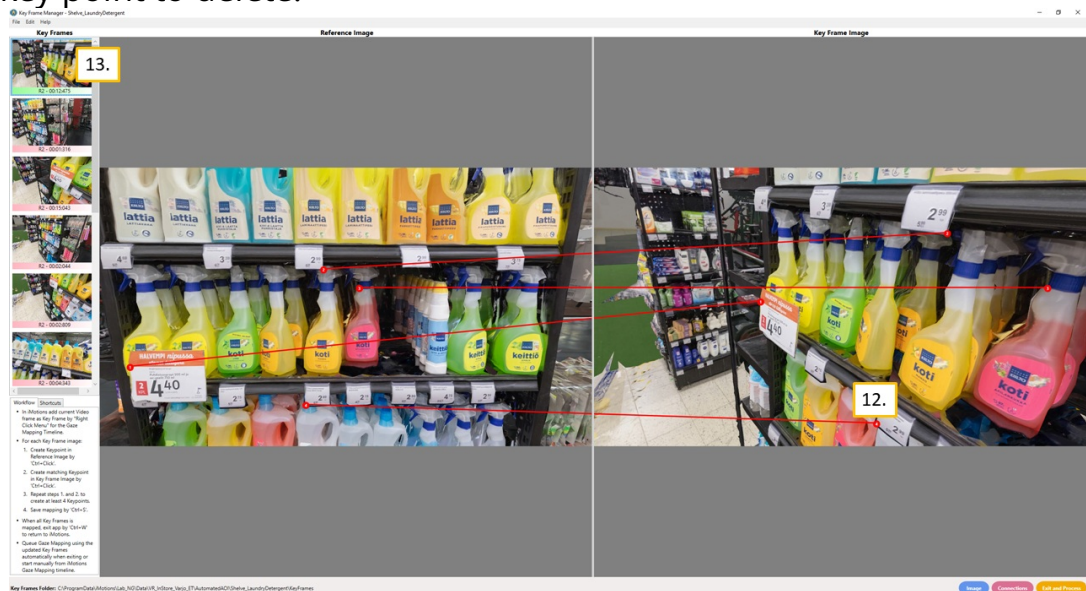


5. If you find a suitable frame, right-click in the area on the timeline that corresponds to the Gaze mapping you are currently correcting.
6. Note the red symbol 'k' on the timeline.
7. *(Repeat the process to place as many keyframes as you like)*



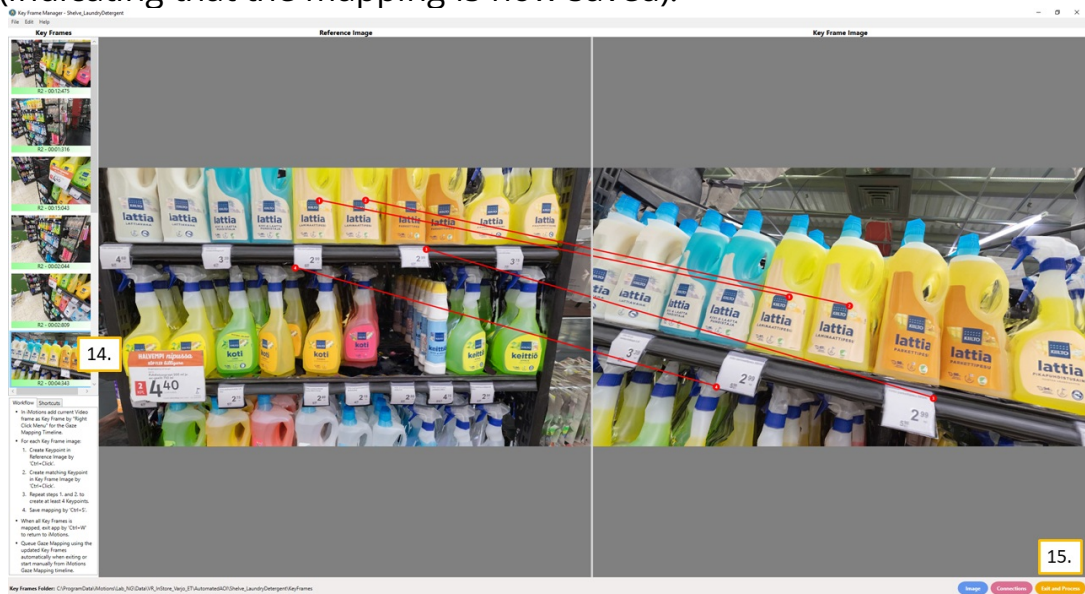
8. When you've placed the keyframes, right-click again in the area on the timeline that corresponds to the Gaze mapping you are currently correcting, and click Open key frame manager

9. Note the list of keyframes on the left side of the Keyframe manager; the bar below the keyframes is colored red. Also note that, for the currently selected keyframe, the reference image is shown side-by-side with the keyframe.
10. Hold down 'CTRL' and click on a feature in the reference image. We recommend picking points that are clearly defined by a corner or other sharp edge.
11. To zoom in, you can use the magnifier by holding down 'SHIFT'. To zoom in, you can use the scrolling wheel.
12. Then, click on the corresponding feature in the keyframe to map the location. Note that, if you misplaced the key point, you can simply drag the key point to a different location. Alternatively, you could also delete the key point, by holding down 'CTRL' while clicking the key point to delete.



13. *(Repeat the process to place at least 4 mapping points per keyframe)*

14. When finished, save the keyframes by pressing 'CTRL + S' on the keyboard; the bar below the selected key frame now colors green (indicating that the mapping is now saved).



15. *(Repeat the process until all frames are mapped)*
16. Click Exit and Process in the bottom-right corner of the Keyframe manager.

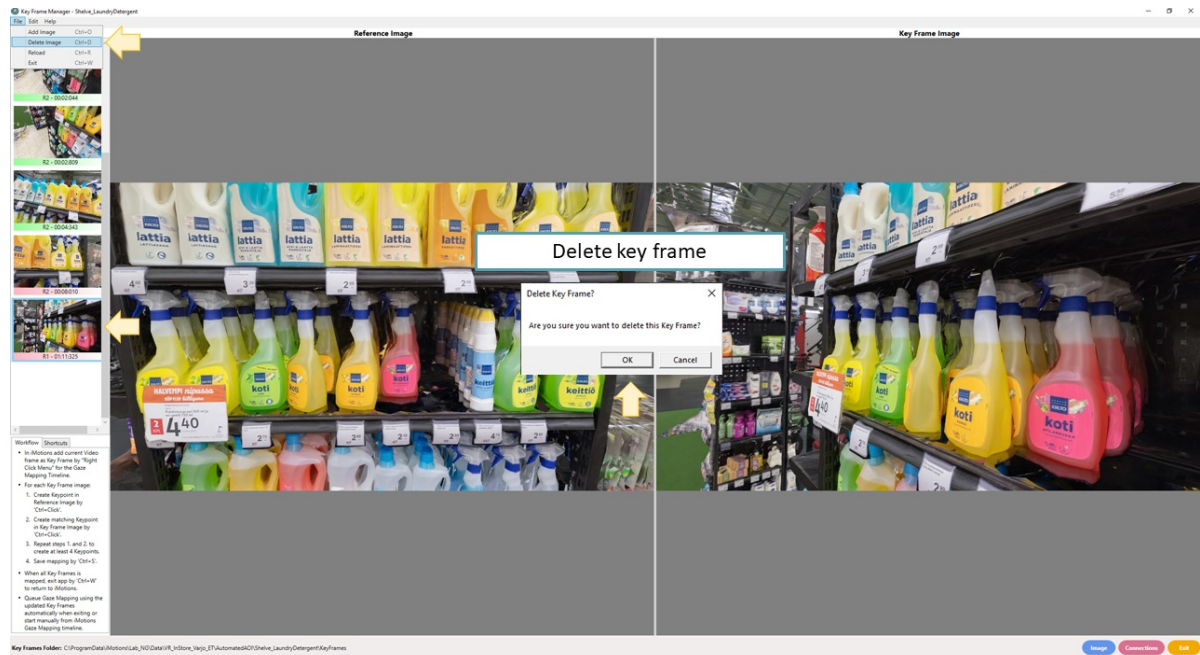
The Gaze mapping algorithm will now repeat the Gaze mapping, taking into consideration the additional information provided by the keyframes.

Important to note: Keyframes are used to “fill gaps” in the gaze mapping data. Therefore, “normal” gaze mapping takes precedence over “key frame” mapping, meaning that adding keyframes will only help to add new gaze mapping to regions in which normal gaze mapping has not been able to process; keyframes cannot be used to change existing gaze mapping, in the form of correction or otherwise. So if you find your gaze mapping to be incorrect, you need to delete the wrongly mapped period first before placing keyframes and restarting the algorithm.



The result after adding keyframes

Note that, if necessary, you can delete keyframes from within the key frame manager.



How to delete a keyframe

To delete a keyframe:

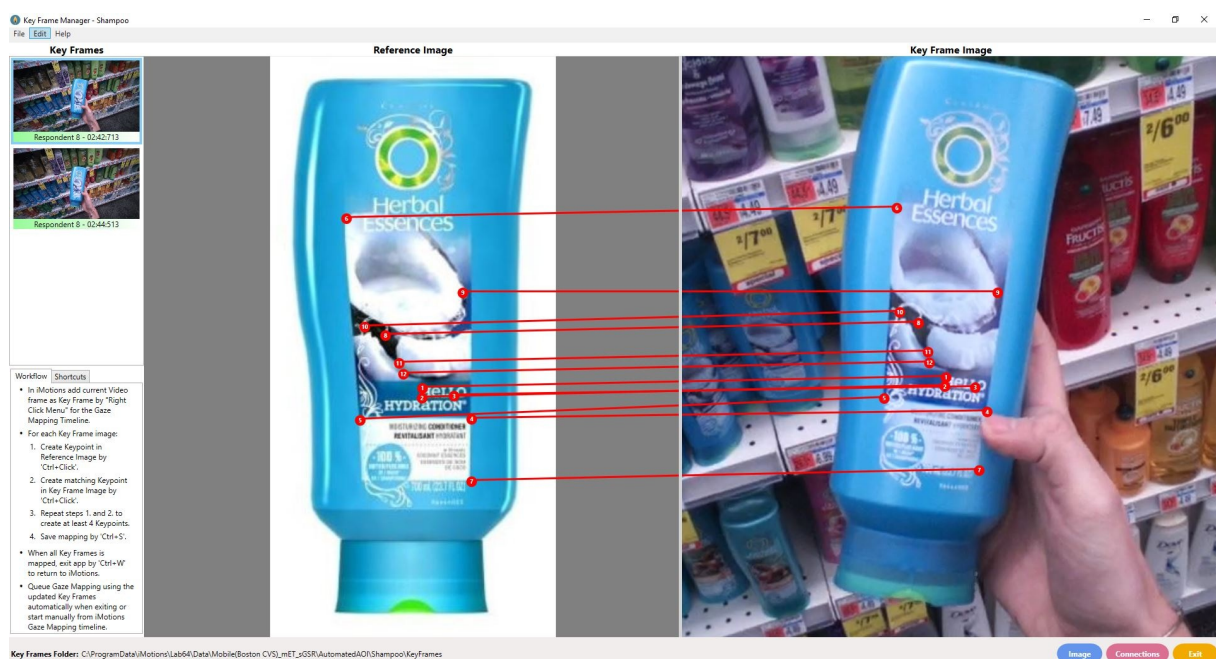
1. Select a keyframe in the list
2. Click File
3. Then, click Delete image
4. Confirm by clicking 'OK'

This is an overview of all shortcuts in the Key Frame Manager:

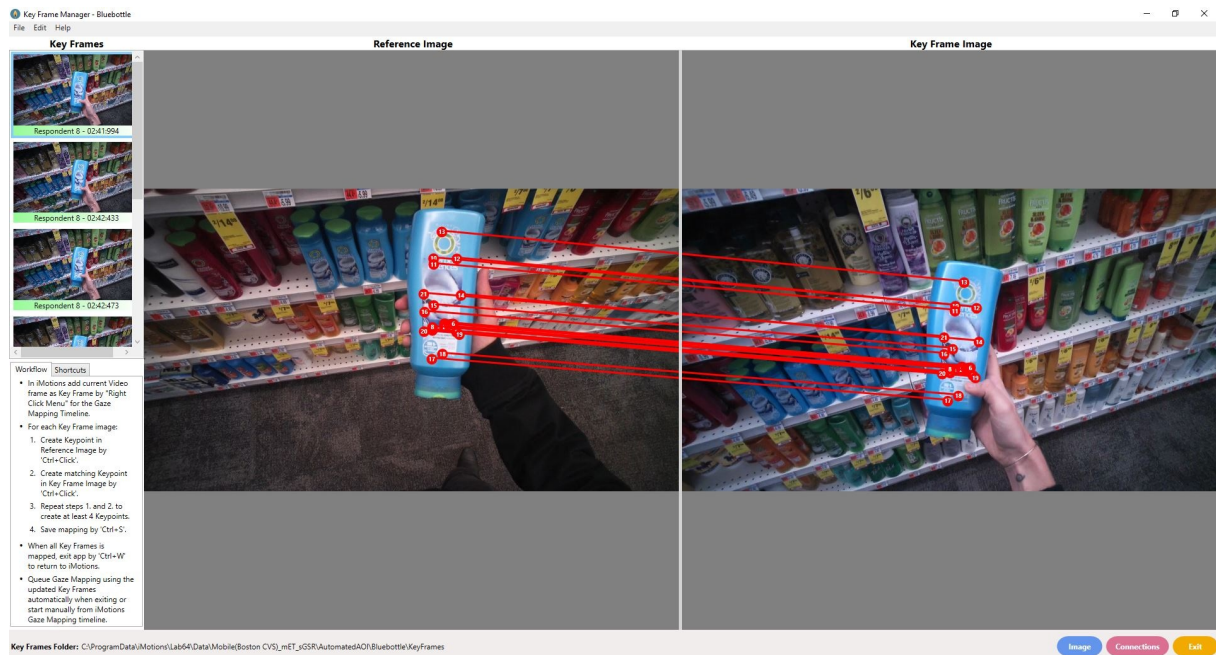
Mouse wheel	Image zoom in/out
Mouse hold and drag the image	Image scroll
Mouse hold and drag point	Point move
Shift	Show magnifier
1	Switch the connections layer visibility
2	Switch the image layer visibility
Ctrl + A	Displays information about the app
Ctrl + O	Open and add new keyframe
Ctrl + R	Reloads current keyframe mapping
Ctrl + D	Deletes currently selected keyframe
Ctrl + S	Save
Ctrl + W	Exit

Examples of when Key Frames can support automated Gaze Mapping

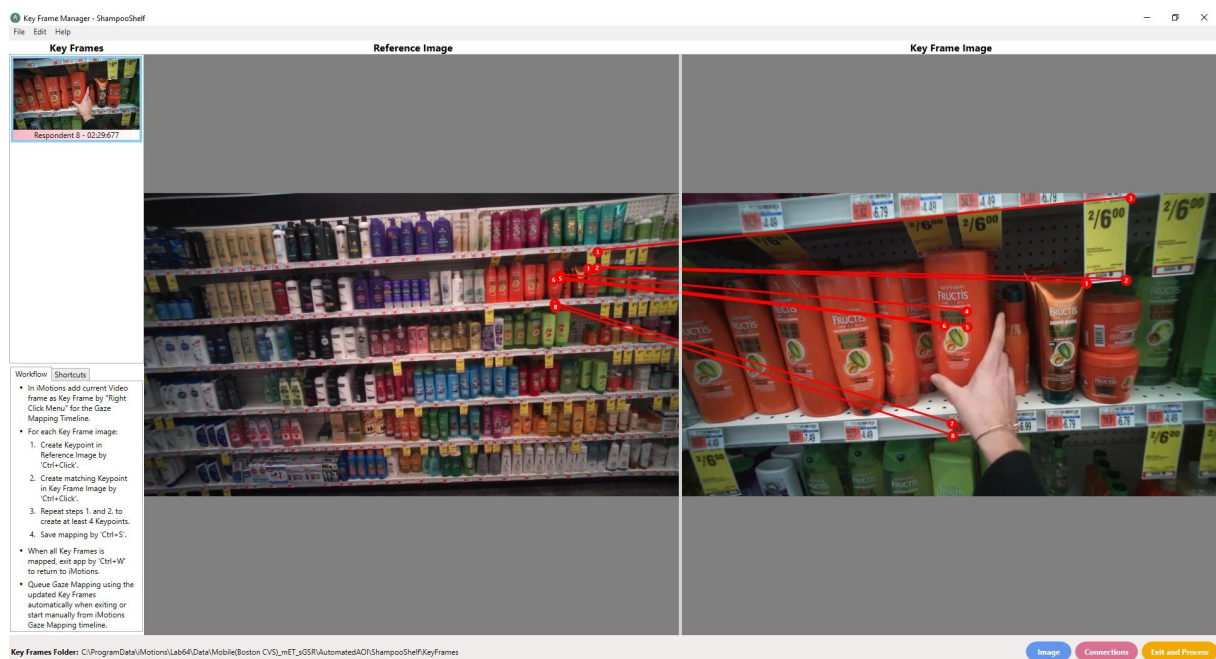
1. Gaze Mapping onto a standalone reference image with very different/missing context information



2. Background variations between the reference image and the original recording

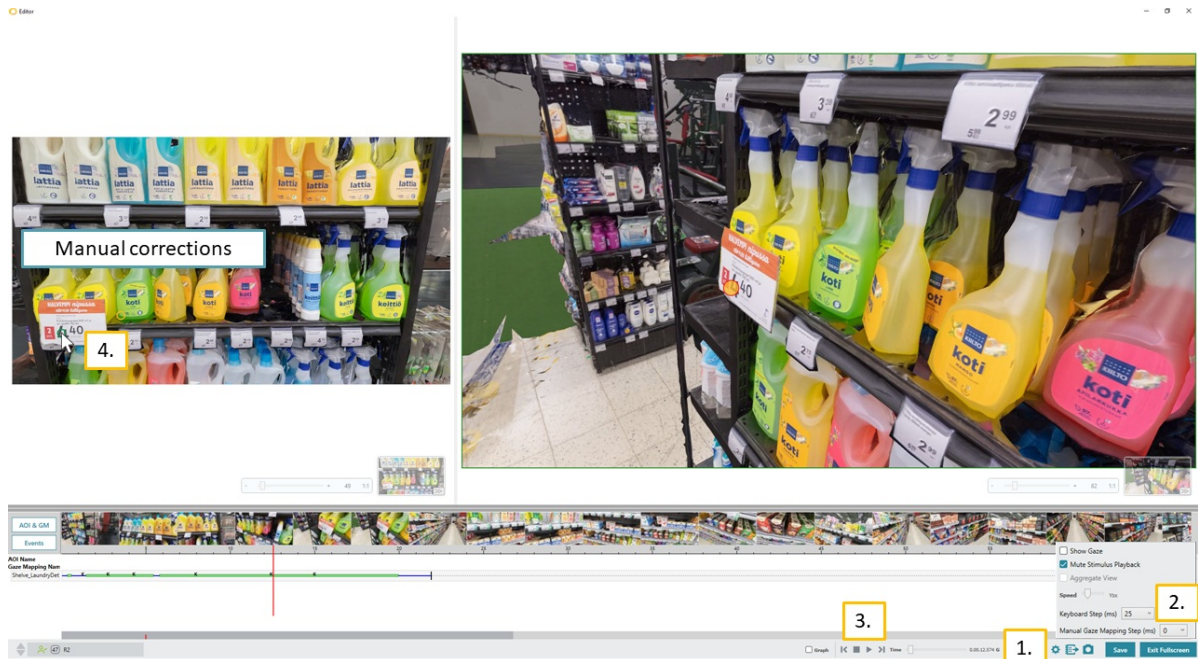


3. Strong zoom leads to loss of context information



Manual corrections

If adding keyframes does not yet lead to a satisfying result, a final option is to review the gaze mapping frame-by-frame and correct misplaced gaze points. This option is mostly relevant if the 'reference' image was successfully identified by the Gaze mapping algorithm, but (some of) the gaze points are still mapped inaccurately.



How to make manual corrections

To manually correct inaccurate gaze points (yellow path):

1. Use the Ctrl + left/right arrow keys on your keyboard to step gaze-by-gaze through the recording. This is advisable when reviewing gaze mapping. If done otherwise, you might miss (or skip over) some gaze mappings!
2. Review the correspondence between the gaze-points in the original eye-tracking recording and the mapped gaze-points in the 'reference' image. Alternatively, use the Previous/Next frame buttons on the timeline.
3. *If the gaze-point is inaccurate*, correct the gaze-point by clicking in the correct location of the 'reference' image (i.e., the location where the gaze-point should have been placed). Importantly, you will not notice at this point that anything happened (!). This is because the underlying algorithm immediately jumps to the next frame to review the next gaze point.

Now proceed in the same way through the entire gaze mapping, and then return to the start to review the corrections. For corrected frames, you will notice the presence of a yellow and a green gaze point in the 'reference' image; the yellow gaze point is the gaze point mapped by the algorithm, while the green gaze point is the corrected gaze point.



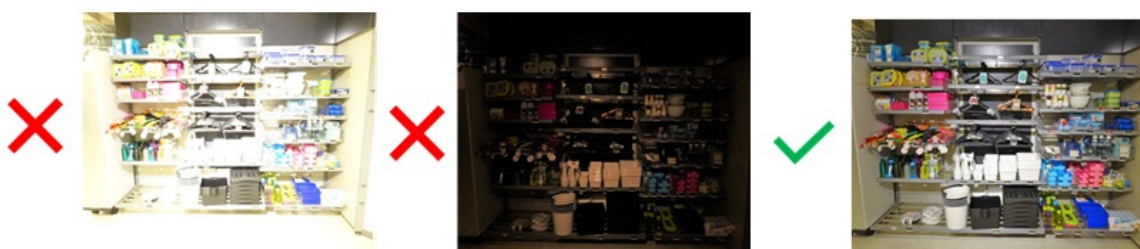
The result of a manual correction

Manual corrections and automatic gaze mappings can be edited as follows: Open the gaze mapping editor, select a gaze mapping, and click the timeline where you have a successful mapping between video and reference image. Here, you click Shift+Ctrl so the reference image will darken a bit to indicate edit mode. While Shift+Ctrl is pressed down and you click a circle in the reference image (either orange/automated or green/manual mapping) it will be removed.

Recommendations for good-quality reference images

Since Gaze Mapping is based on computer vision, the quality of the reference image will impact the success of the Gaze Mapping procedure.

The ideal reference image is well-lit, not warped, and sharp. Clear contrasts and visible details help the algorithm to detect all the objects well. Also, make sure that the picture is neither overexposed (too bright) nor underexposed (too dark). Likewise, if the recording environment is too bright or too dark, it will be difficult for the algorithm to perform the Gaze Mapping.



If the real-world object is difficult to capture in a single picture (like a very long shopping aisle) you can take multiple pictures and then merge them in an external software. Please make sure that these merged images line up well.

*If you want to see Gaze mapping in action in a relevant demo study, or if you want to play around with the Gaze mapping routine yourself, please consider downloading the **In-store VR (Shelf-testing in VR environment)** demo study.*